

Digital Design & Computer Arch.

Lecture 10: Microarchitecture Fundamentals & Design

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Agenda for Today & Next Few Lectures

- Instruction Set Architectures (ISA): LC-3 and MIPS
- Assembly programming: LC-3 and MIPS
- Microarchitecture (principles & single-cycle uarch)
- Multi-cycle microarchitecture
- Pipelining
- Issues in Pipelining:
 - Control & Data Dependence Handling
 - State Maintenance and Recovery
- Out-of-Order Execution

Problem
Algorithm
Program/Language
System Software
SW/HW Interface
Micro-architecture
Logic
Devices
Electrons

Readings

■ This week

- Introduction to microarchitecture and single-cycle microarchitecture
 - H&H, Chapter 7.1-7.3
 - P&P, Appendices A and C
- Multi-cycle microarchitecture
 - H&H, Chapter 7.4
 - P&P, Appendices A and C

■ Next week

- Pipelining
 - H&H, Chapter 7.5
- Pipelining Issues
 - H&H, Chapter 7.7, 7.8.1-7.8.3

Recall: Microarchitecture

- **Implementation of the ISA** under specific **design constraints and goals**
- Anything done in hardware without exposure to software
 - Pipelining
 - In-order versus out-of-order instruction execution
 - Memory access scheduling policy
 - Speculative execution
 - Superscalar processing (multiple instruction issue?)
 - Clock gating
 - Caching? Levels, size, associativity, replacement policy
 - Prefetching?
 - Voltage/frequency scaling?
 - Error correction?

Implementing the ISA: Microarchitecture Basics

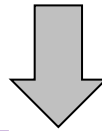
Now That We Know What an ISA Is...

- How do we implement it?
- i.e., **how do we design a system that obeys the hardware/software interface?**
- Aside: “System” can be solely hardware or a combination of hardware and software
 - Recall the “Translation of ISAs”
 - An **ISA** can be converted (by software or hardware) into an **implementation ISA**
- We will assume “completely hardware” implementation for most lectures

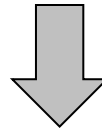
How Does a Machine Process Instructions?

- What does processing an instruction mean?
- We will assume the von Neumann model (for now)

AS = Architectural (programmer visible) state before an instruction is processed



Process instruction



AS' = Architectural (programmer visible) state after an instruction is processed

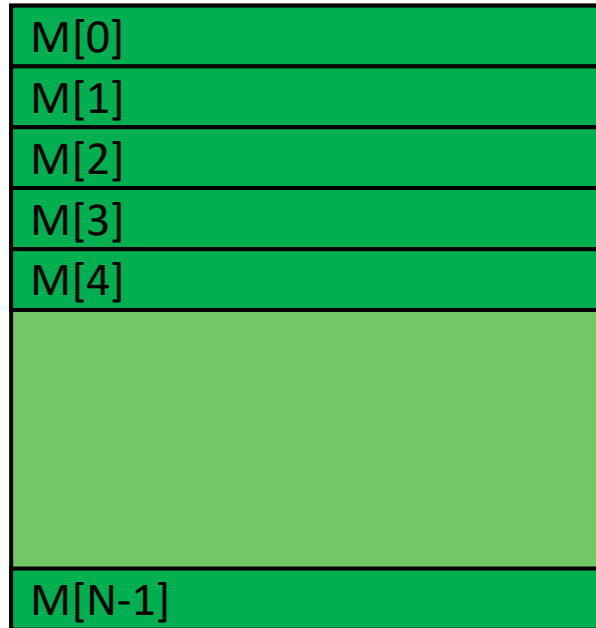
- Processing an instruction: Transforming AS to AS' according to the ISA specification of the instruction

Recall: von Neumann Model/Architecture

Stored program

Sequential instruction processing

Recall: Programmer Visible (Architectural) State



Memory

array of storage locations
indexed by an address



Registers

- given special names in the ISA (as opposed to addresses)
- general vs. special purpose

Program Counter

memory address
of the current (or next) instruction

Instructions (and programs) specify how to transform
the values of programmer visible state

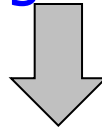
The “Process Instruction” Step

- ISA specifies abstractly what AS' should be, given an instruction and AS
 - It defines an **abstract finite state machine** where
 - State = programmer-visible state
 - Next-state logic = instruction execution specification
 - From ISA point of view, there are no “intermediate states” between AS and AS' during instruction execution
 - One state transition per instruction
- Microarchitecture implements how AS is transformed to AS'
 - There are many choices in implementation
 - We can have programmer-invisible state to optimize the speed of instruction execution: **multiple** state transitions per instruction
 - Choice 1: $AS \rightarrow AS'$ (transform AS to AS' in a single clock cycle)
 - Choice 2: $AS \rightarrow AS+MS1 \rightarrow AS+MS2 \rightarrow AS+MS3 \rightarrow AS'$ (take multiple clock cycles to transform AS to AS')

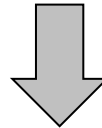
A Very Basic Instruction Processing Engine

- Each instruction takes a single clock cycle to execute
- Only combinational logic is used to implement instruction execution
 - *No intermediate, programmer-invisible state updates*

AS = Architectural (programmer visible) state
at the beginning of a clock cycle



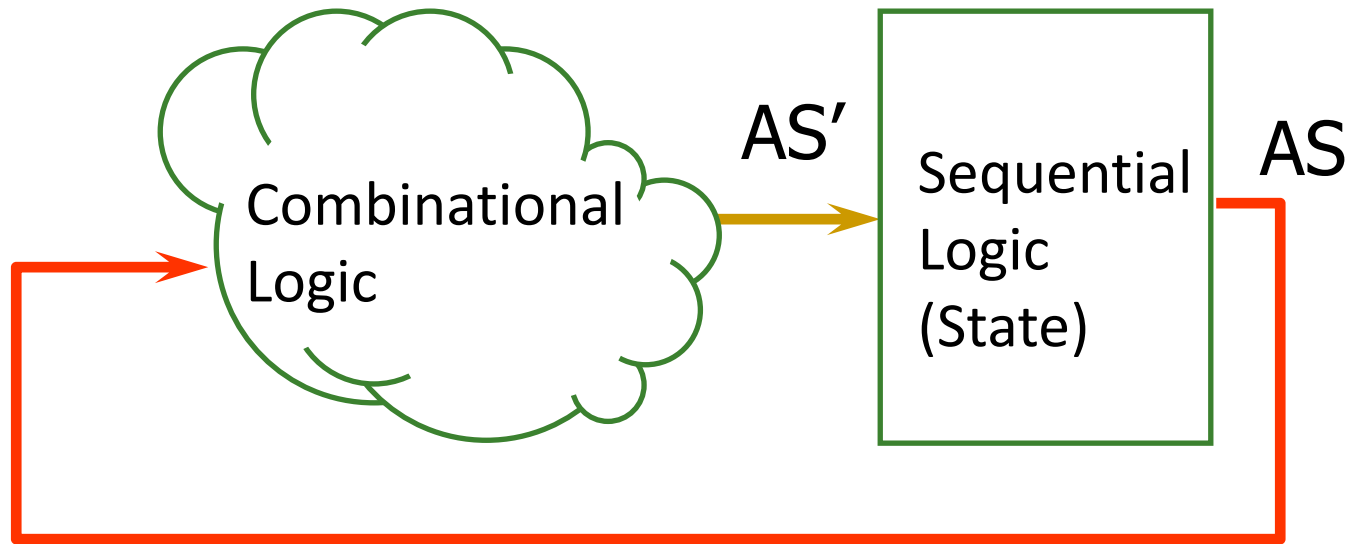
Process instruction in **one clock cycle**



AS' = Architectural (programmer visible) state
at the end of a clock cycle

A Very Basic Instruction Processing Engine

- Single-cycle machine



- What is the *clock cycle time* determined by?
- What is the *critical path* (i.e., longest delay path) of the combinational logic determined by?

Single-cycle vs. Multi-cycle Machines

■ Single-cycle machines

- ❑ Each instruction takes a single clock cycle
- ❑ All state updates made at the end of an instruction's execution
- ❑ Big disadvantage: The slowest instruction determines cycle time → long clock cycle time

■ Multi-cycle machines

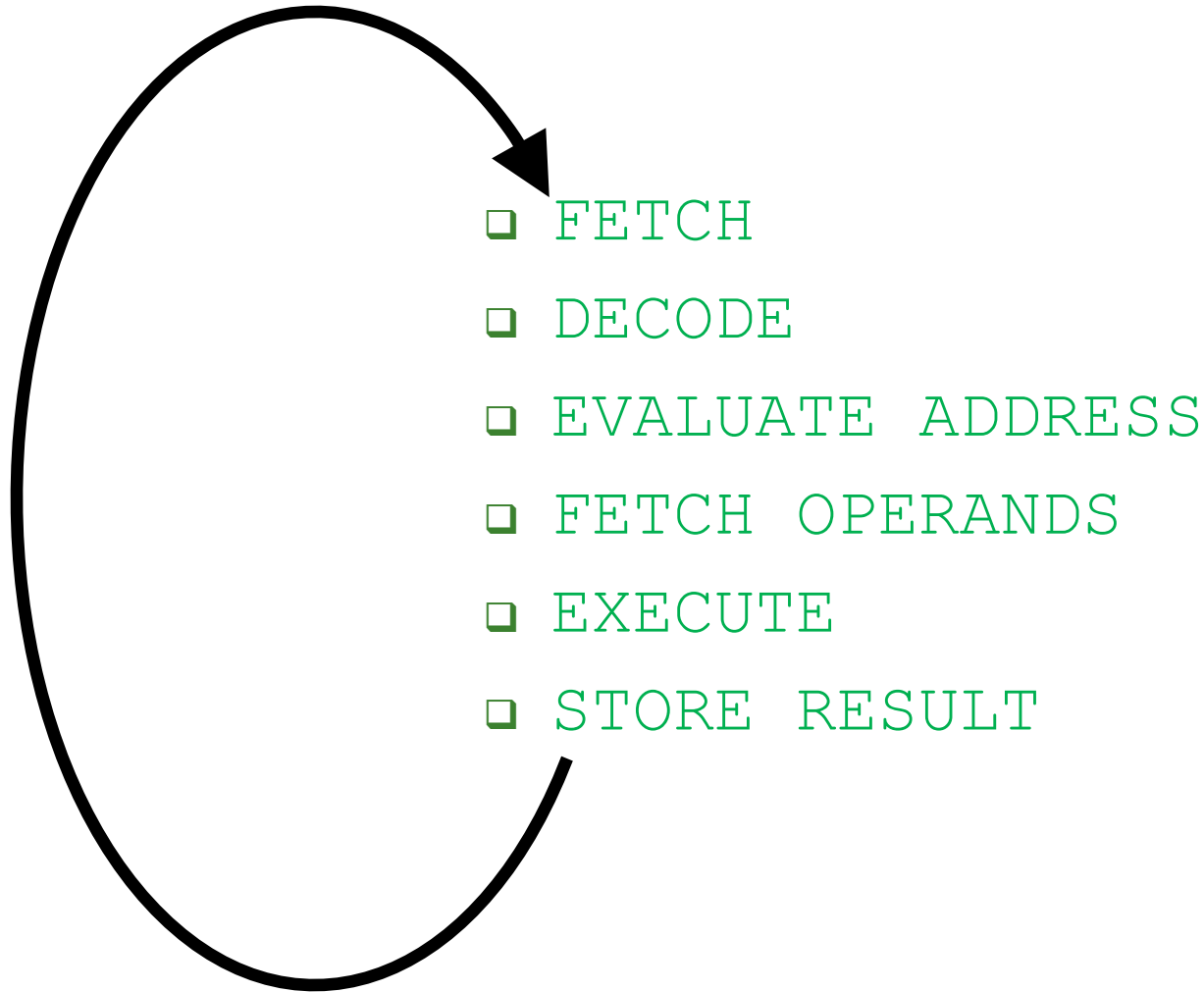
- ❑ Instruction processing broken into multiple cycles/stages
- ❑ State updates can be made during an instruction's execution
- ❑ Architectural state updates made at the end of an instruction's execution
- ❑ Advantage over single-cycle: The slowest "stage" determines cycle time

- Both single-cycle and multi-cycle machines literally follow the von Neumann model at the microarchitecture level

Instruction Processing “Cycle”

- Instructions are processed under the direction of a “control unit” step by step.
- Instruction cycle: Sequence of steps to process an instruction
- Fundamentally, there are six steps:
 - Fetch
 - Decode
 - Evaluate Address
 - Fetch Operands
 - Execute
 - Store Result
- Not all instructions require all six steps (see P&P Ch. 4)

Recall: The Instruction Processing “Cycle”



Instruction Processing “Cycle” vs. Machine Clock Cycle

- Single-cycle machine:
 - All six phases of the instruction processing cycle take a *single machine clock cycle* to complete
- Multi-cycle machine:
 - All six phases of the instruction processing cycle can take *multiple machine clock cycles* to complete
 - In fact, **each phase can take multiple clock cycles to complete**

Instruction Processing Viewed Another Way

- Instructions transform Data (AS) to Data' (AS')
- This transformation is done by functional units
 - Units that “operate” on data
- These units need to be told what to do to the data
- An instruction processing engine consists of two components
 - **Datapath**: Consists of **hardware elements that deal with and transform data signals**
 - **functional units** that operate on data
 - **hardware structures** (e.g., wires, muxes, decoders, tri-state bufs) that enable the flow of data into the functional units and registers
 - **storage units** that store data (e.g., registers)
 - **Control logic**: Consists of **hardware elements that determine control signals, i.e., signals that specify what the datapath elements should do to the data**

Recall: LC-3: A von Neumann Machine

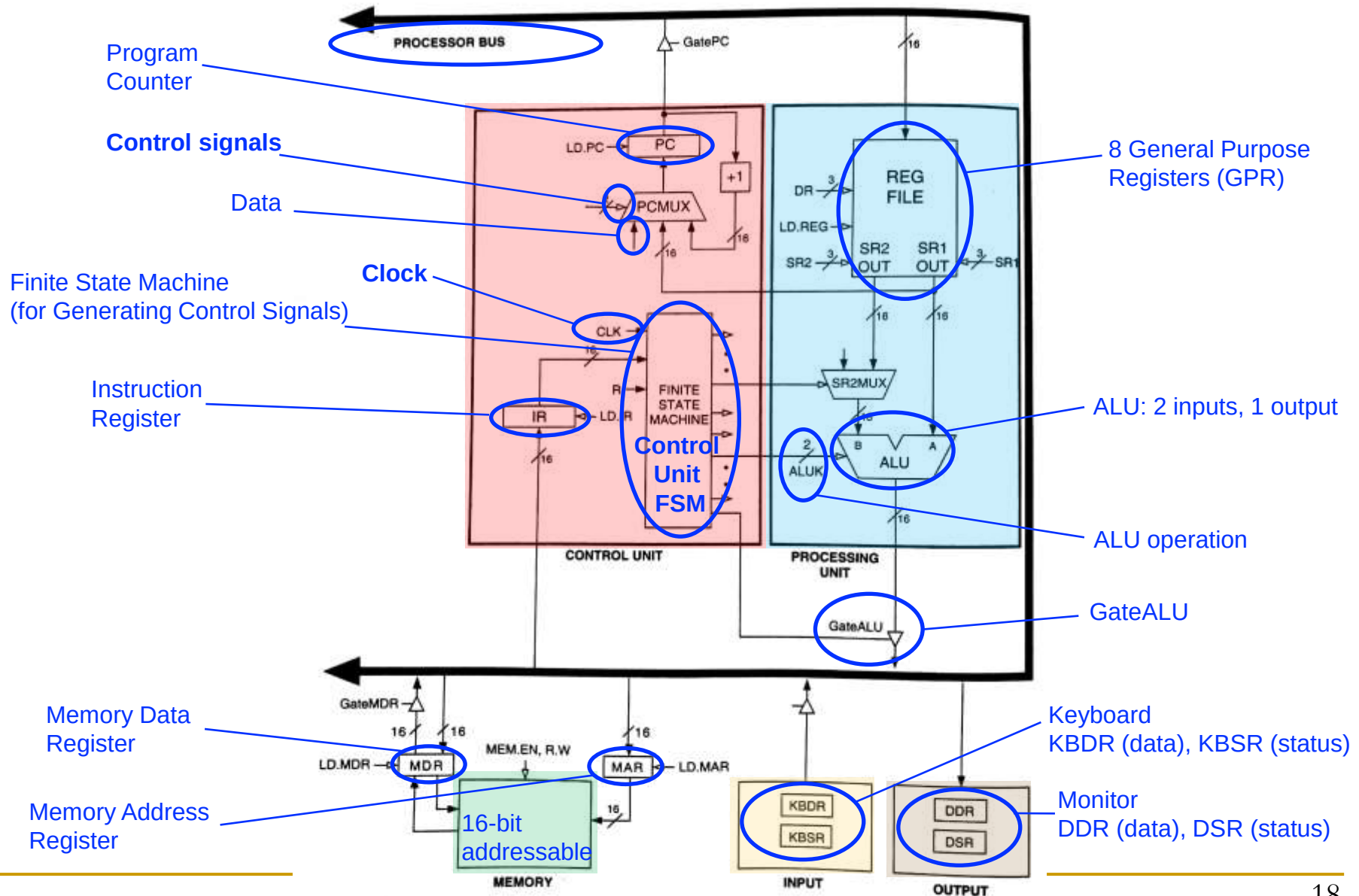


Figure 4.3 The LC-3 as an example of the von Neumann model

Single-cycle vs. Multi-cycle: Control & Data

- Single-cycle machine:
 - Control signals are generated in the same clock cycle as the one during which data signals are operated on
 - Everything related to an instruction happens in one clock cycle (serialized processing)
- Multi-cycle machine:
 - Control signals needed in the next cycle can be generated in the current cycle
 - Latency of control processing can be overlapped with latency of datapath operation (more parallelism)
- See P&P Appendix C for more (microprogrammed multi-cycle microarchitecture)

Many Ways of Datapath and Control Design

- There are many ways of designing the datapath and control logic
- Example ways
 - Single-cycle, multi-cycle, pipelined, out-of-order datapath & control
 - Single-bus vs. multi-bus datapaths
 - Hardwired/combinational vs. microcoded/microprogrammed control
 - Control signals generated by combinational logic versus
 - Control signals stored in a memory structure
- Control signals and structure depend on the datapath design

Flash-Forward: Performance Analysis

- Execution time of a single instruction

- **{CPI} x {clock cycle time}** CPI: Cycles Per Instruction

- Execution time of an entire program

- Sum over all instructions [**{CPI} x {clock cycle time}**]
- **{# of instructions} x {Average CPI} x {clock cycle time}**

- Single-cycle microarchitecture performance

- $\text{CPI} = 1$
- Clock cycle time = long

- Multi-cycle microarchitecture performance

- CPI = different for each instruction
 - Average CPI $\rightarrow$ hopefully small
- Clock cycle time = short

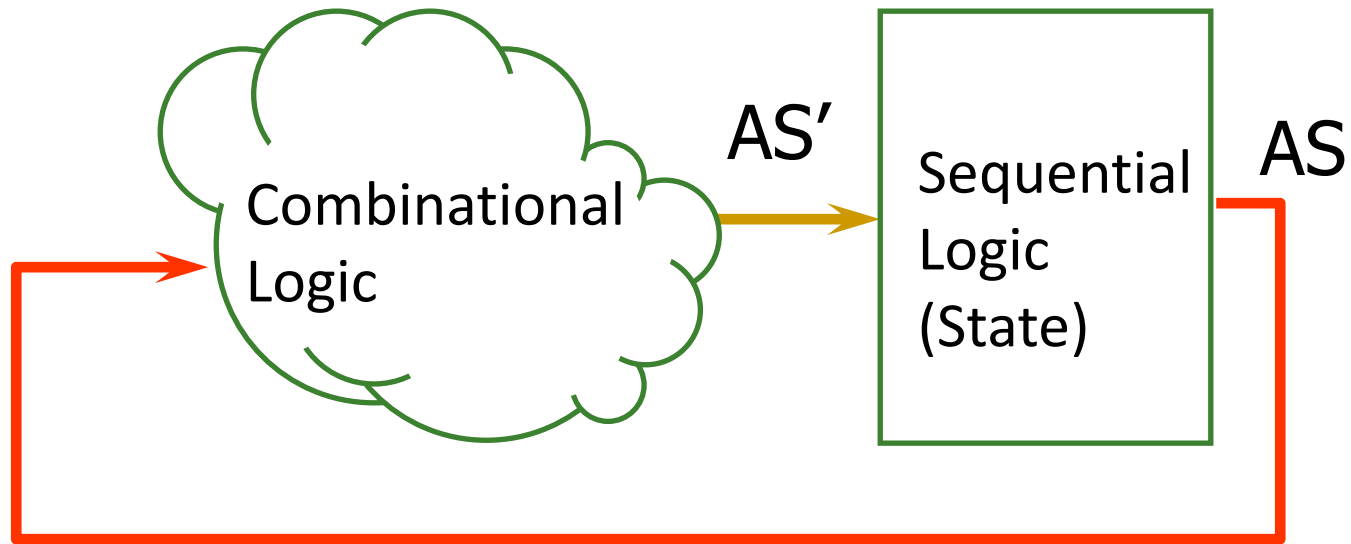
In multi-cycle, we have two degrees of freedom to optimize independently

A Single-Cycle Microarchitecture

From the Ground Up

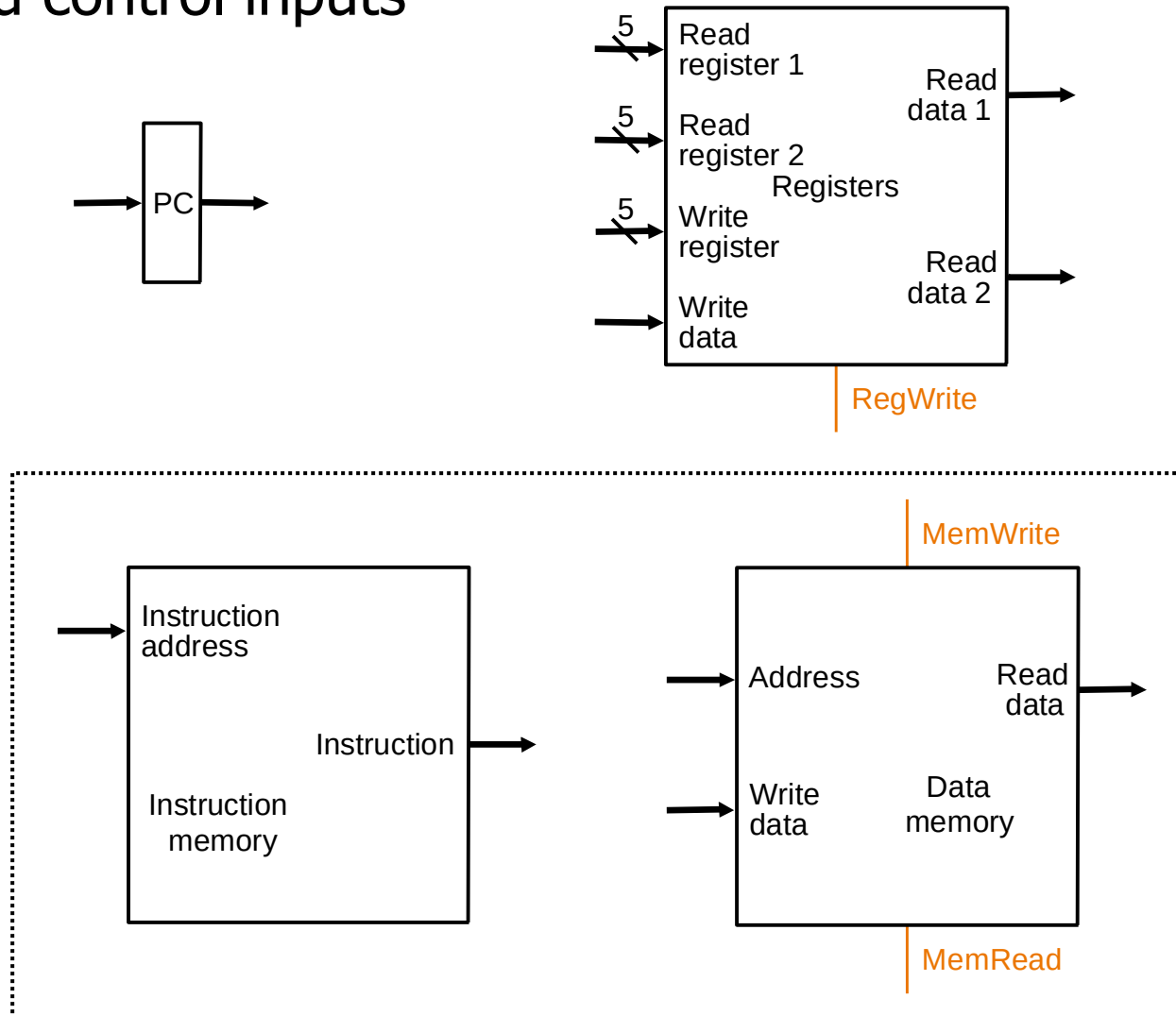
Remember...

- Single-cycle machine

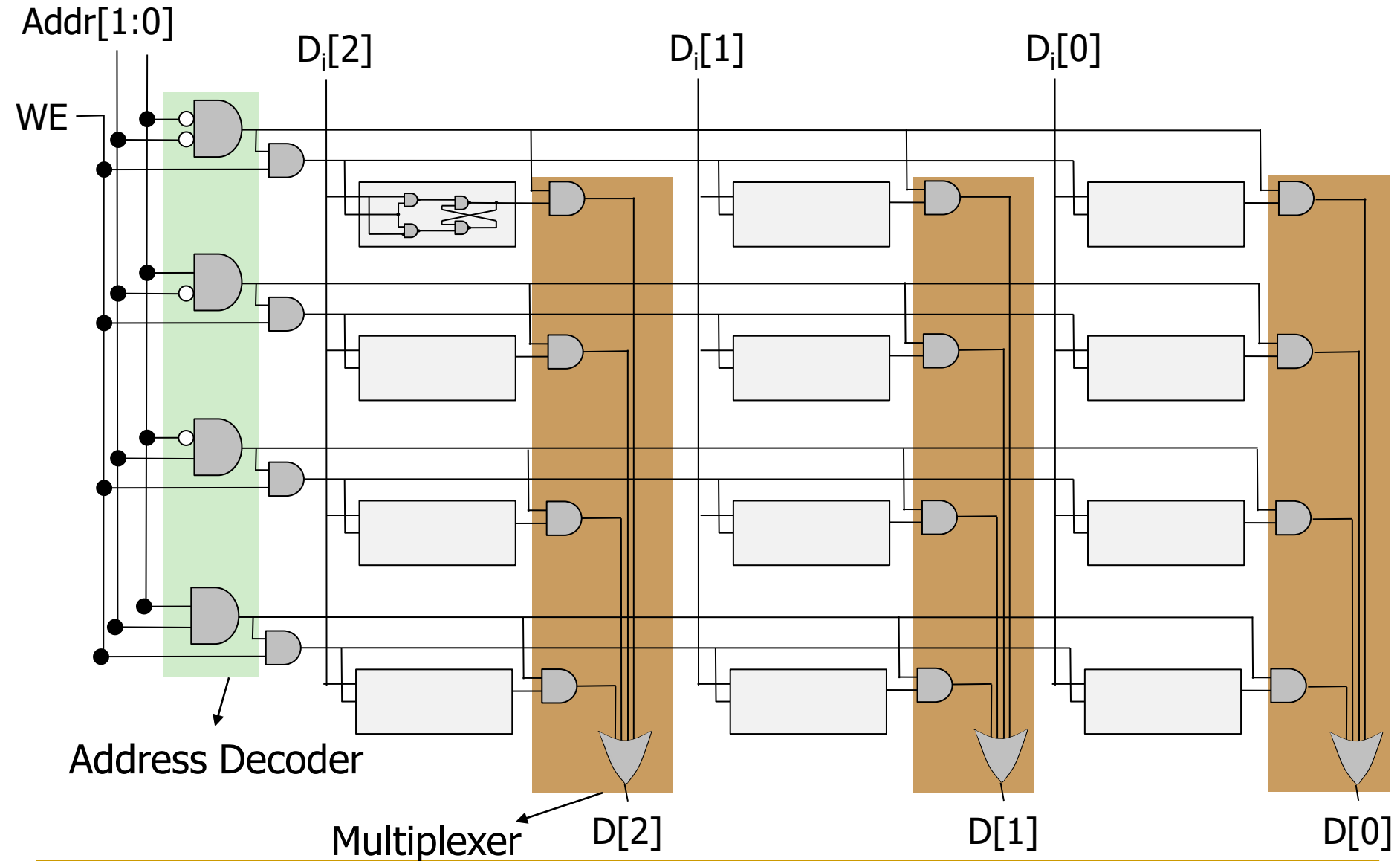


Let's Start with the State Elements (MIPS)

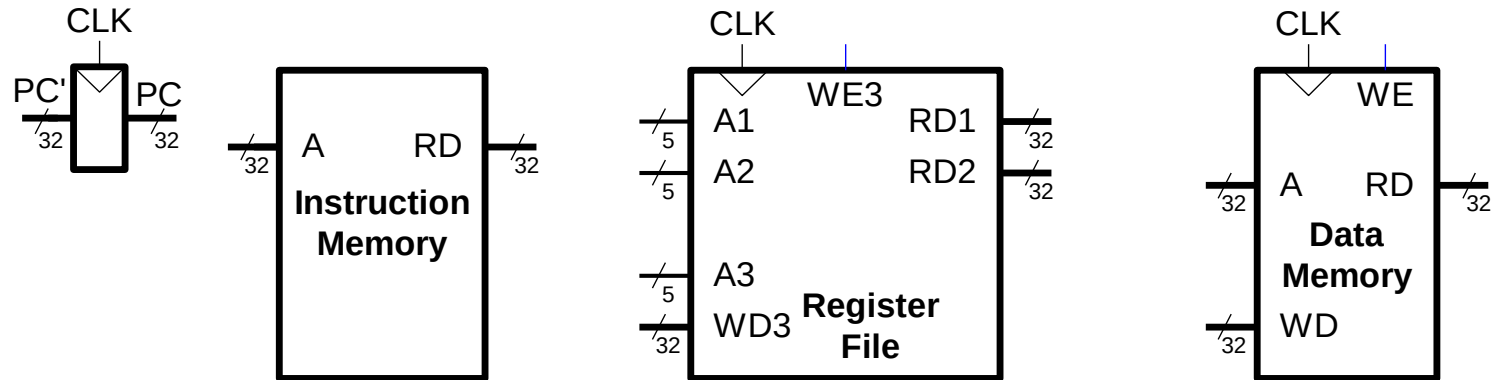
■ Data and control inputs



Recall: A Memory Array (4 locations X 3 bits)



MIPS State Elements



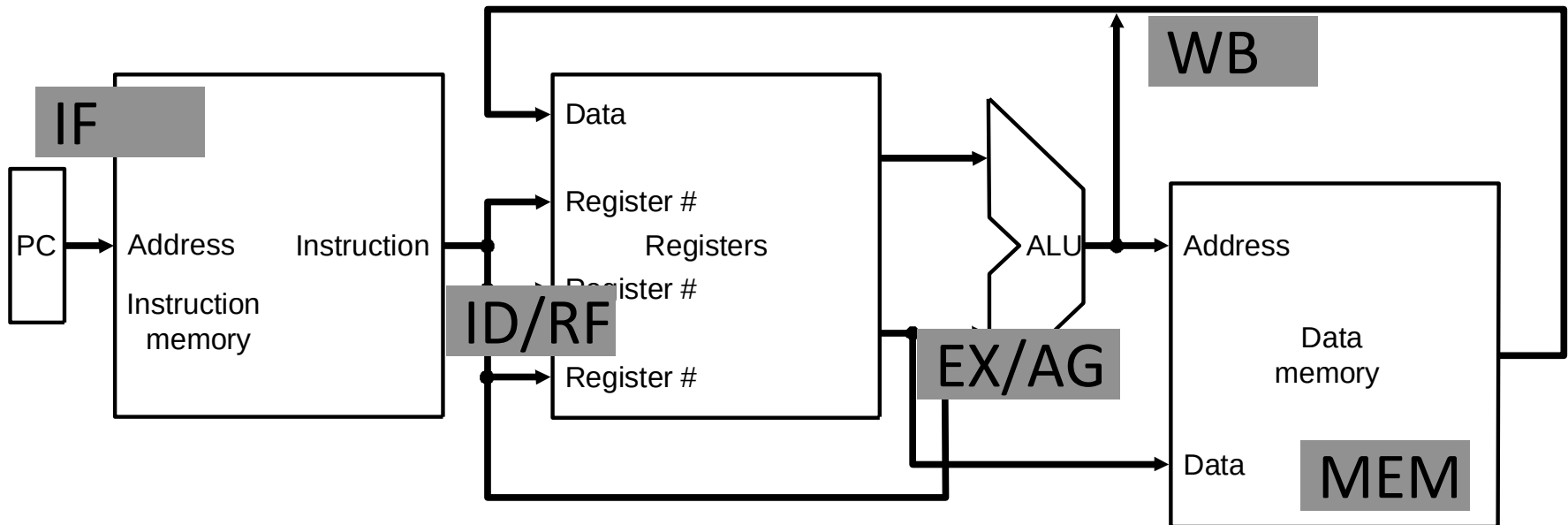
- ❑ **Program counter:**
32-bit register
- ❑ **Instruction memory:**
Takes input 32-bit address A and reads the 32-bit data (i.e., instruction) from that address to the read data output RD
- ❑ **Register file:**
The 32-element, 32-bit register file has 2 read ports and 1 write port
- ❑ **Data memory:**
If the write enable, WE, is 1, it writes 32-bit data WD into memory location at 32-bit address A on the rising edge of the clock.
If the write enable is 0, it reads 32-bit data from address A onto RD.

For Now, We Will Assume

- Ultra-fast (i.e., single-cycle) memory and register file
- Combinational read
 - output of the read data port is a combinational function of the memory contents and the corresponding input address
- Synchronous write
 - target location is updated at the positive edge clock transition when the write enable signal is asserted
 - Cannot affect read output in-between positive clock edges
- Single-cycle memory
 - Contrast this with memory that tells when the data is ready
 - i.e., Ready signal: indicating the read or write is done
 - See P&P Appendix C (LC3-b) for multi-cycle memory

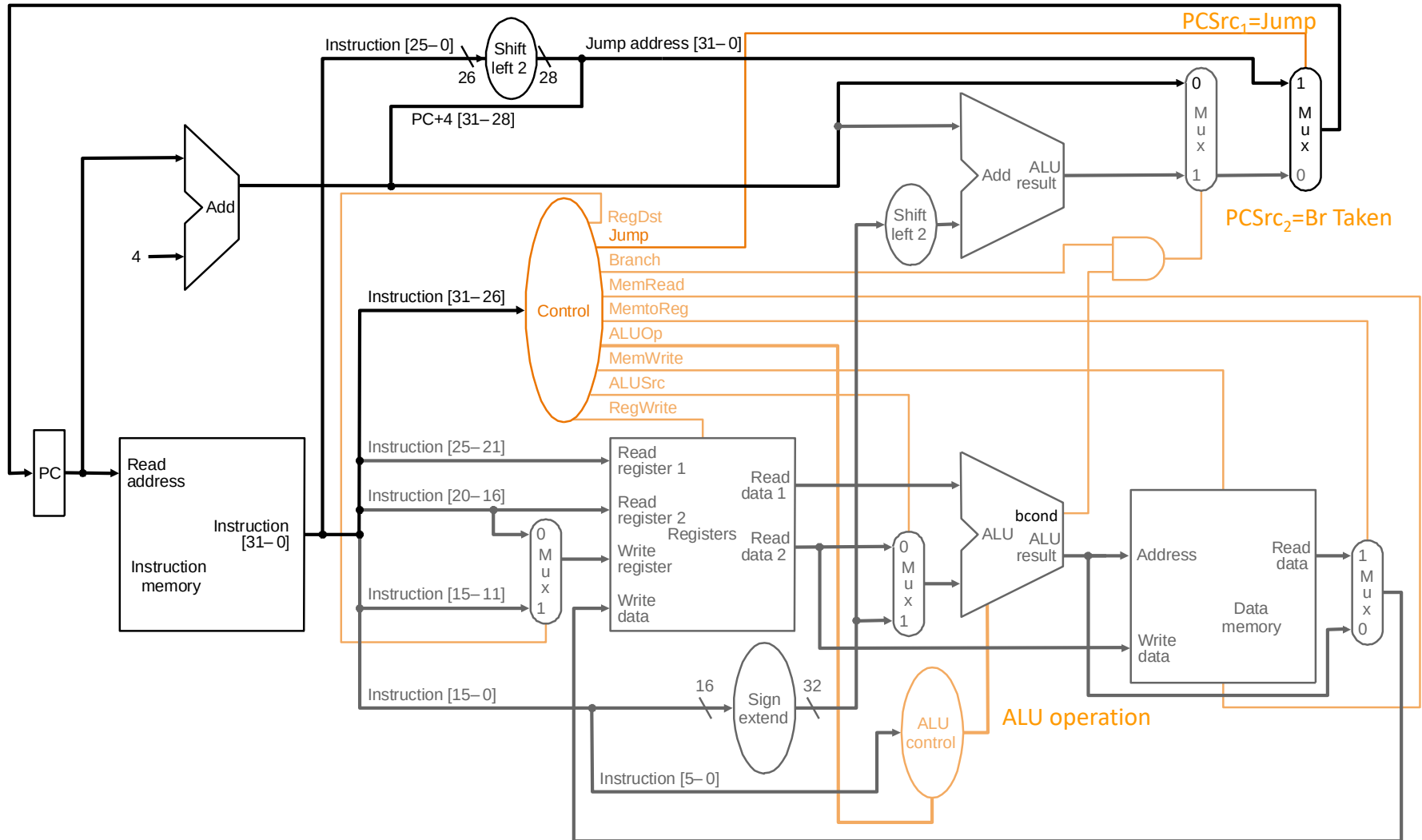
Instruction Processing

- 5 generic steps
 - ❑ Instruction fetch (IF)
 - ❑ Instruction decode and register operand fetch (ID/RF)
 - ❑ Execute/Evaluate memory address (EX/AG)
 - ❑ Memory operand fetch (MEM)
 - ❑ Store/writeback result (WB)

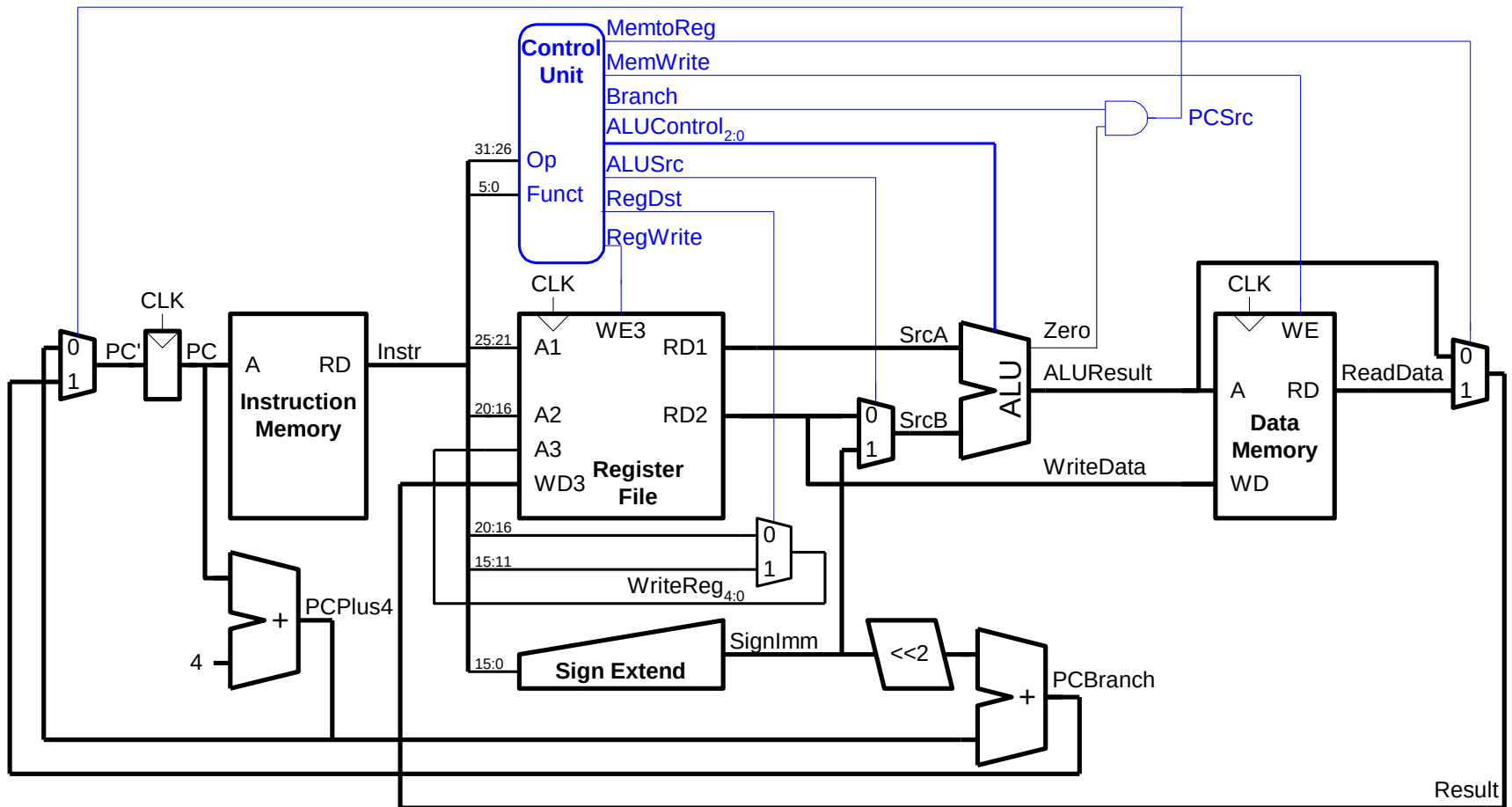


We Need to Provide the
Datapath & Control Logic
to Execute All ISA Instructions

What Is To Come: Single-Cycle MIPS Processor



Another Complete Single-Cycle Processor



Single-Cycle Datapath for *Arithmetic and Logical Instructions*

R-Type ALU Instructions

- R-type: 3 register operands

MIPS assembly (e.g., register-register signed addition)

```
add    $s0, $s1, $s2           # $s0=rd, $s1=rs, $s2=rt
```

Machine Encoding

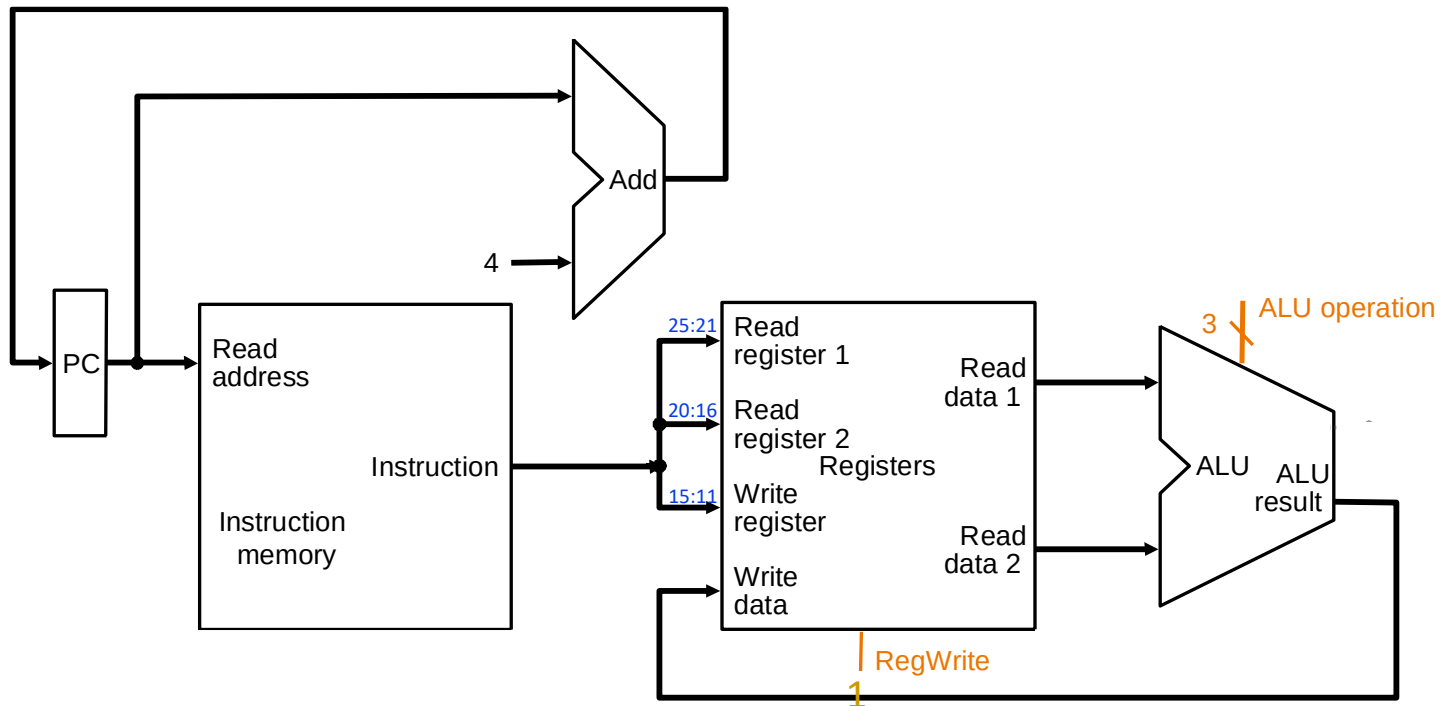
0	rs	rt	rd	0	add (32)
6 bits	5 bits	5 bits	5 bits	5 bits	6 bits

R-Type

- Semantics

```
if MEM[PC] == add rd rs rt
    GPR[rd] ← GPR[rs] + GPR[rt]
    PC ← PC + 4
```

(R-Type) ALU Datapath



IF	ID	EX	MEM	WB
----	----	----	-----	----

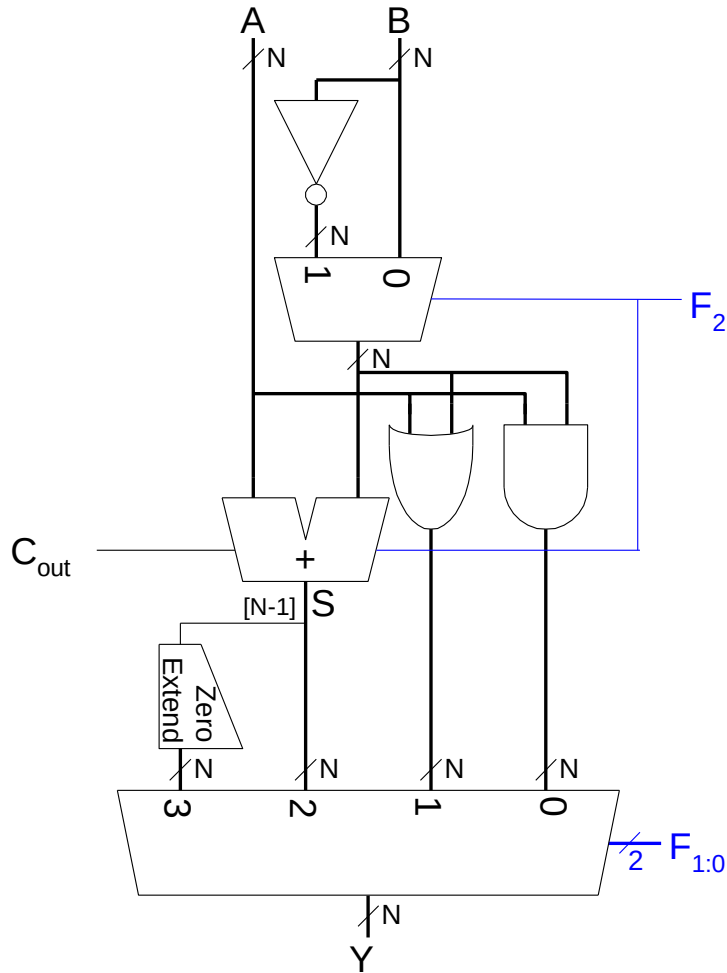


Combinational
state update logic

if MEM[PC] == ADD rd rs rt
GPR[rd] $\leftarrow$ GPR[rs] + GPR[rt]
PC $\leftarrow$ PC + 4

Example: ALU Design

- **ALU operation** ($F_{2:0}$) comes from the control logic



$F_{2:0}$	Function
000	$A \& B$
001	$A \mid B$
010	$A + B$
011	not used
100	$A \& \sim B$
101	$A \mid \sim B$
110	$A - B$
111	SLT

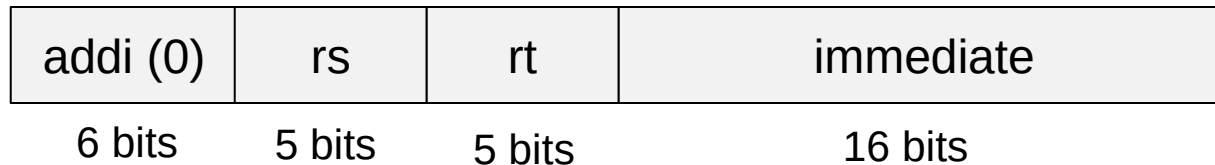
I-Type ALU Instructions

- I-type: 2 register operands and 1 immediate

MIPS assembly (e.g., register-immediate signed addition)

```
addi $s0, $s1, 5           # $s0 = rt, $s1 = rs
```

Machine Encoding



I-Type

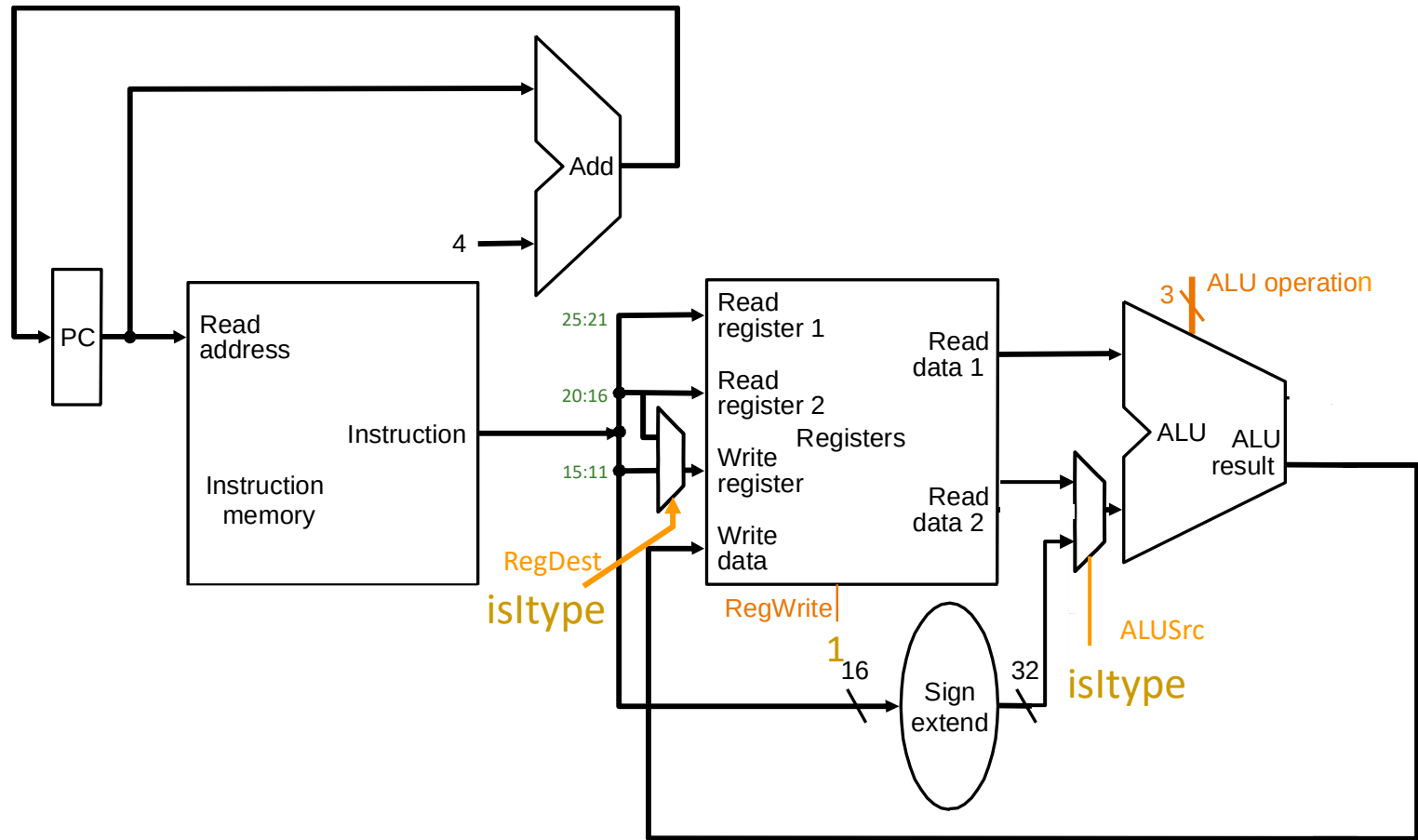
- Semantics

if MEM[PC] == addi rs rt immediate

PC $\leftarrow$ PC + 4

GPR[rt] $\leftarrow$ GPR[rs] + sign-extend(immediate)

Datapath for R- and I-Type ALU Insts.



IF	ID	EX	MEM	WB
----	----	----	-----	----

if MEM[PC] == ADDI rt rs immediate

$GPR[rt] \leftarrow GPR[rs] + \text{sign-extend}(\text{immediate})$

$PC \leftarrow PC + 4$



Combinational
state update logic

Recall: ADD with one Literal in LC-3

■ ADD assembly and machine code

LC-3 assembly

```
ADD R1, R4, #-2
```

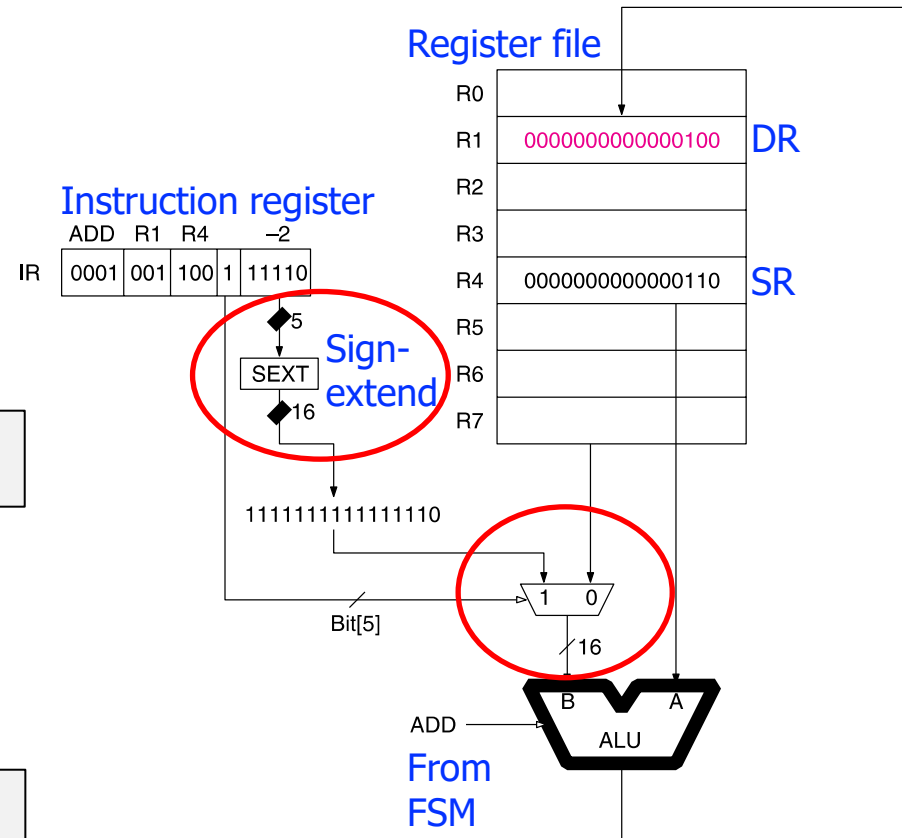
Field Values

OP	DR	SR	imm5
1	1	4	1
-2			

Machine Code

OP	DR	SR	imm5
0001	001	100	1
11110			

15 12 11 9 8 6 5 4 0



Single-Cycle Datapath for *Data Movement Instructions*

Load Instructions

- Load 4-byte word

MIPS assembly

```
lw    $s3, 8($s0)           # $s0=rs, $s3=rt
```

Machine Encoding

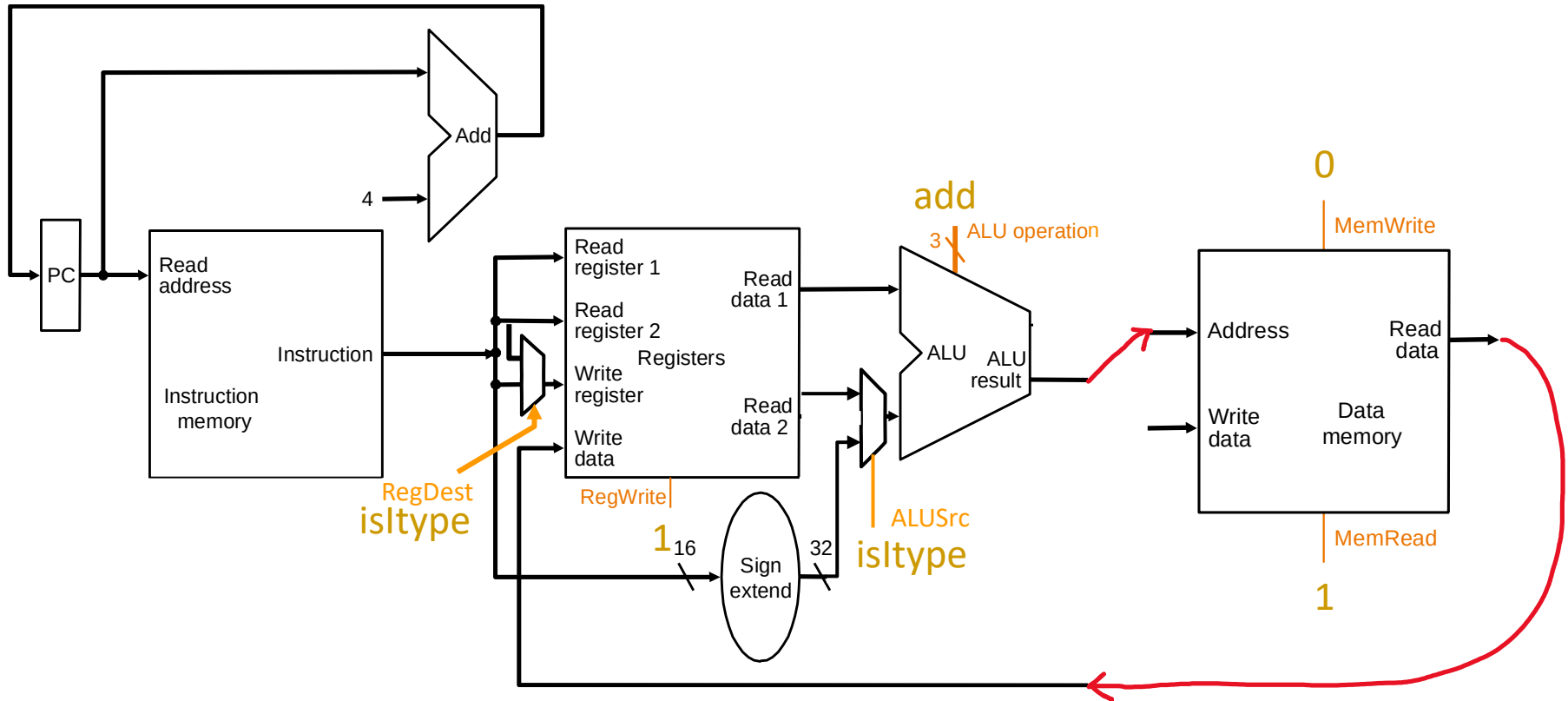


I-Type

- Semantics

```
if MEM[PC] == lw rt offset16 (base)
    PC ← PC + 4
    address = sign-extend(offset) + GPR(base)
    GPR[rt] ← MEM[address]
```


LW Datapath



if $\text{MEM}[\text{PC}] == \text{LW rt offset}_{16} (\text{base})$
 $\text{address} = \text{sign-extend}(\text{offset}) + \text{GPR}[\text{base}]$
 $\text{GPR}[\text{rt}] \leftarrow \text{MEM}[\text{address}]$
 $\text{PC} \leftarrow \text{PC} + 4$

IF	ID	EX	MEM	WB
----	----	----	-----	----

Combinational
state update logic

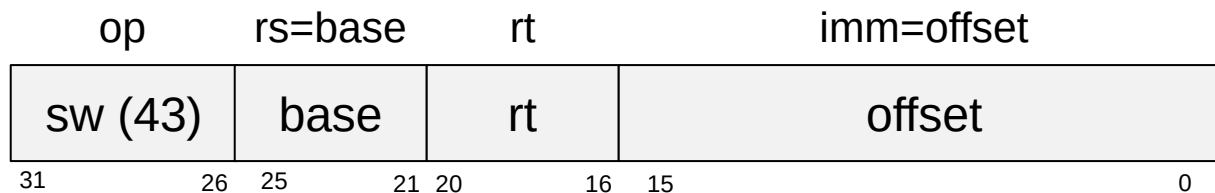
Store Instructions

- Store 4-byte word

MIPS assembly

```
sw    $s3, 8($s0)           # $s0=rs, $s3=rt
```

Machine Encoding



I-Type

- Semantics

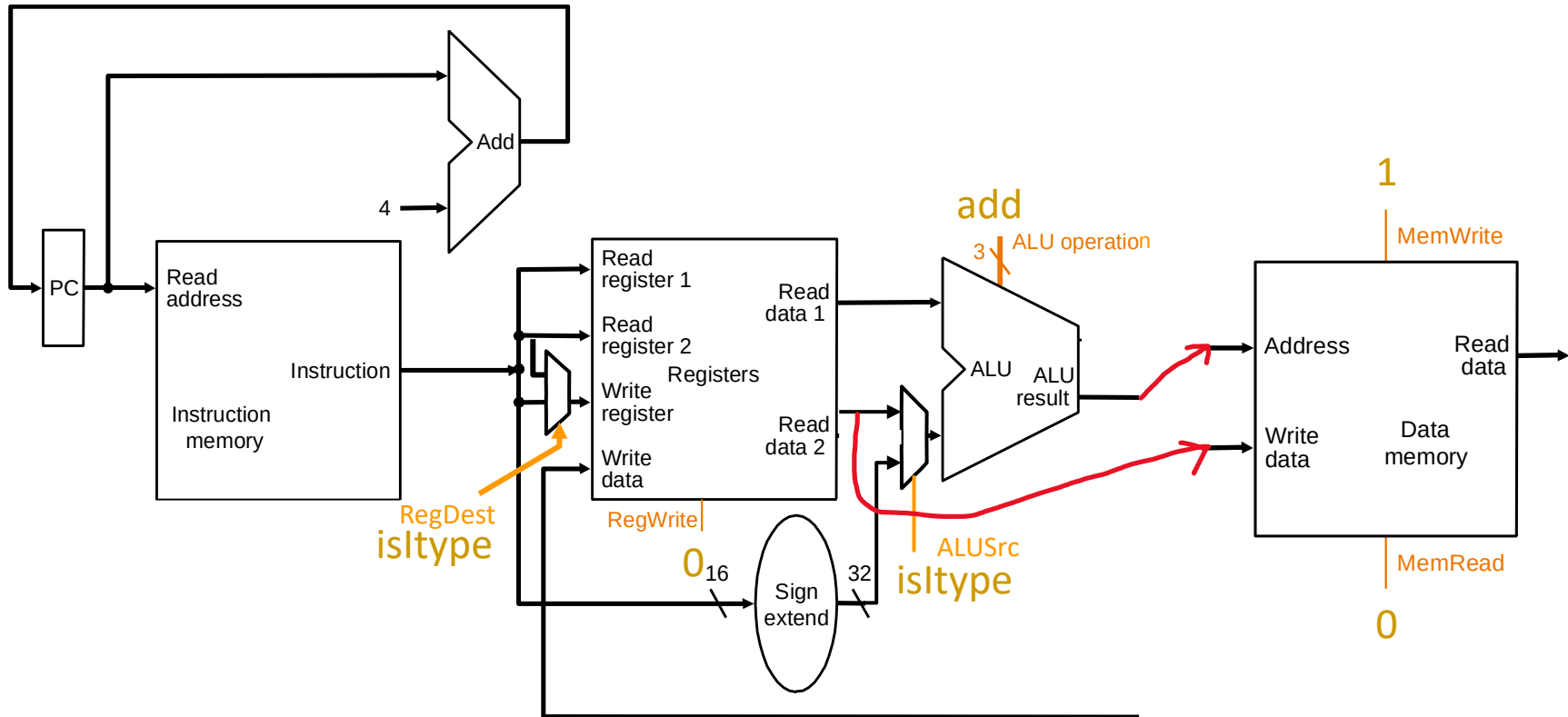
if $\text{Mem}[\text{PC}] == \text{sw rt offset}_{16}(\text{base})$

$\text{PC} \leftarrow \text{PC} + 4$

$\text{address} = \text{sign-extend}(\text{offset}) + \text{GPR}(\text{base})$

$\text{MEM}[\text{address}] \leftarrow \text{GPR}[\text{rt}]$

SW Datapath

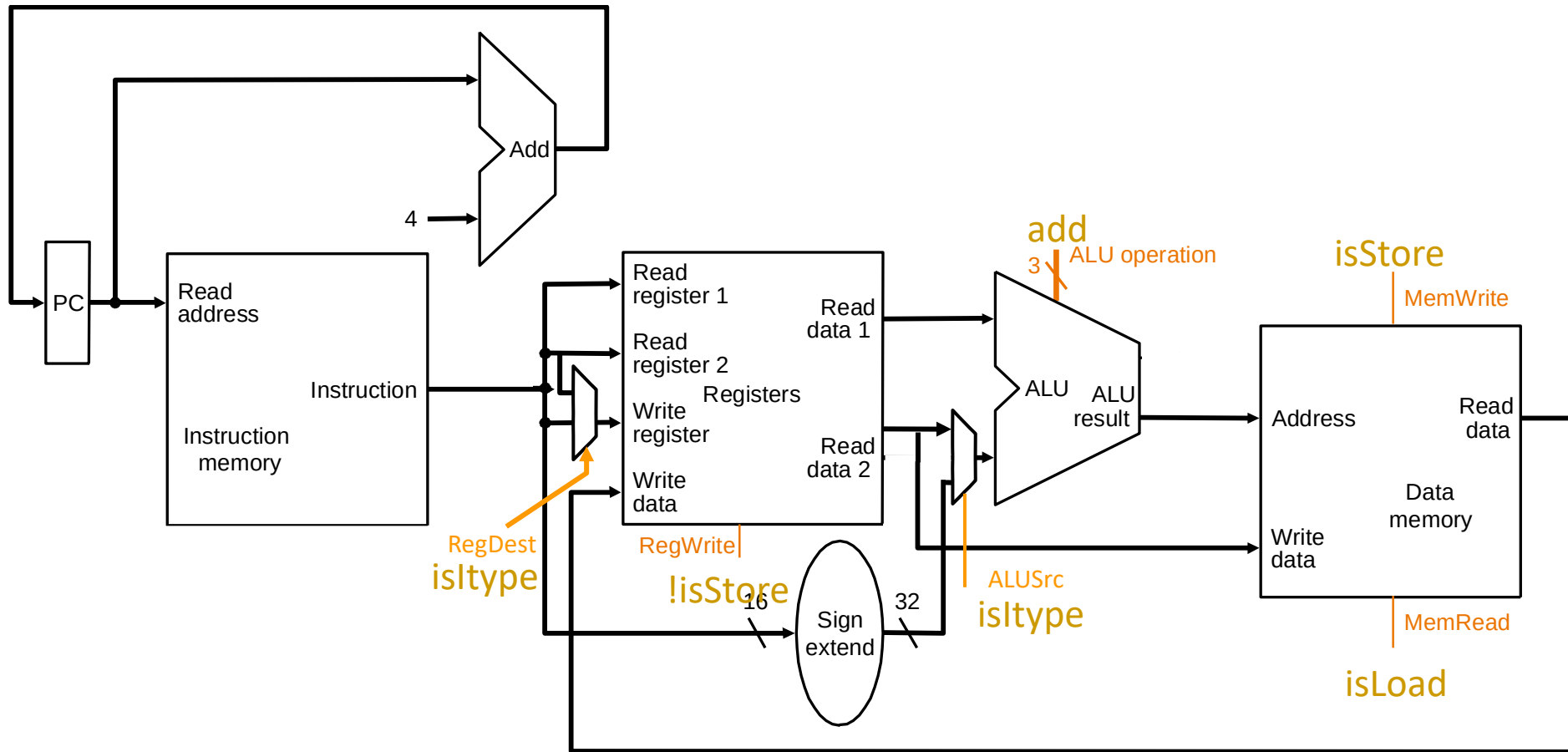


if $\text{MEM}[\text{PC}] == \text{SW rt offset}_{16}(\text{base})$
 $\text{address} = \text{sign-extend}(\text{offset}) + \text{GPR}[\text{base}]$
 $\text{MEM}[\text{address}] \leftarrow \text{GPR}[\text{rt}]$
 $\text{PC} \leftarrow \text{PC} + 4$

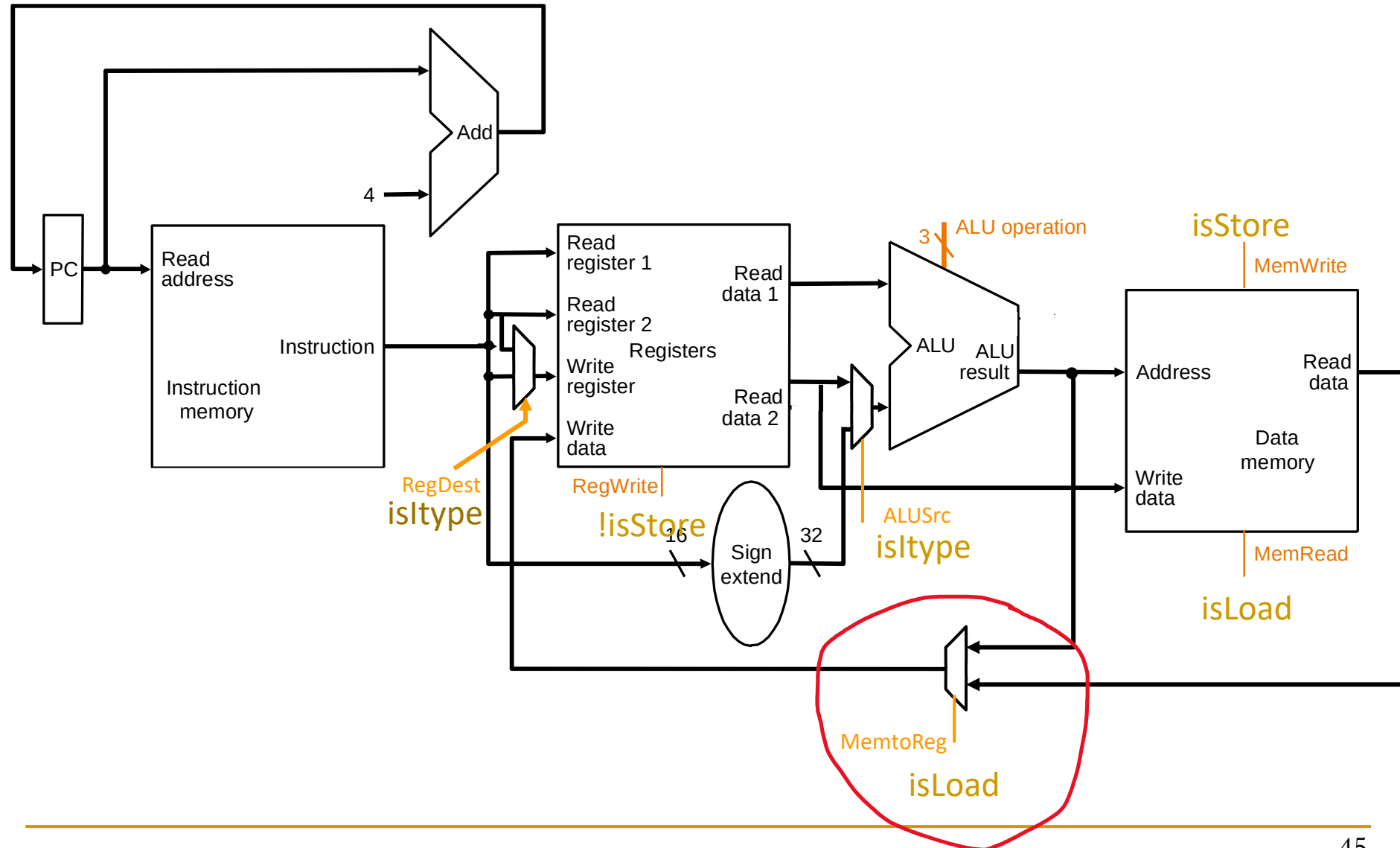
IF	ID	EX	MEM	WB
----	----	----	-----	----

Combinational
state update logic

Load-Store Datapath



Datapath for Non-Control-Flow Insts.

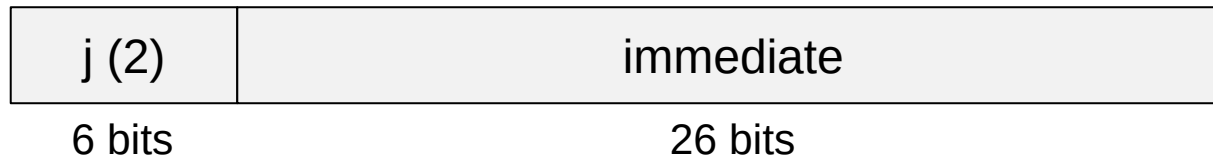


Single-Cycle Datapath for *Control Flow Instructions*

Jump Instruction

- Unconditional branch or jump

j target



J-Type

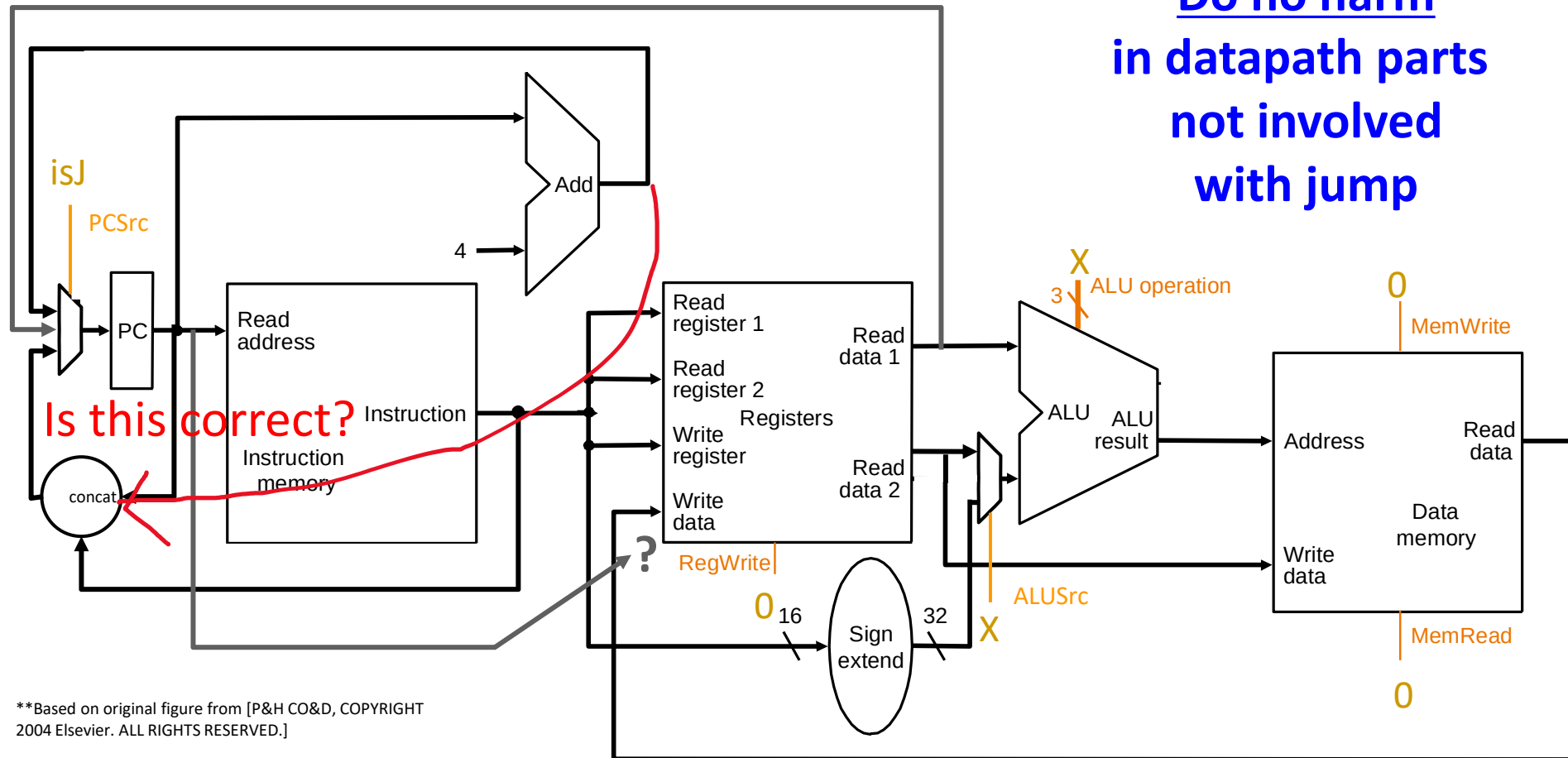
- 2 = opcode
- immediate (target) = target address

- Semantics

if $\text{MEM}[\text{PC}] == j \text{ immediate}_{26}$
 $\text{target} = \{ \text{PC} + [31:28], \text{immediate}_{26}, 2' \text{ b}00 \}$
 $\text{PC} \leftarrow \text{target}$

Unconditional Jump Datapath

Do no harm
in datapath parts
not involved
with jump



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if $MEM[PC] == J \text{ immediate}_{26}$
 $PC = \{ PC \uparrow [31:28], \text{immediate}_{26}, 2' b00 \}$

What about JR, JAL, JALR?

Other Jumps in MIPS

- ❑ **jr: jump register**

Semantics

if $\text{MEM}[\text{PC}] == \text{jr rs}$
 $\text{PC} \leftarrow \text{GPR}(\text{rs})$

- ❑ **jal: jump and link (function calls)**

Semantics

if $\text{MEM}[\text{PC}] == \text{jal immediate}_{26}$
 $\$ra \leftarrow \text{PC} + 4$
 $\text{target} = \{ \text{PC}^\dagger[31:28], \text{immediate}_{26}, 2'b00 \}$
 $\text{PC} \leftarrow \text{target}$

- ❑ **jalr: jump and link register**

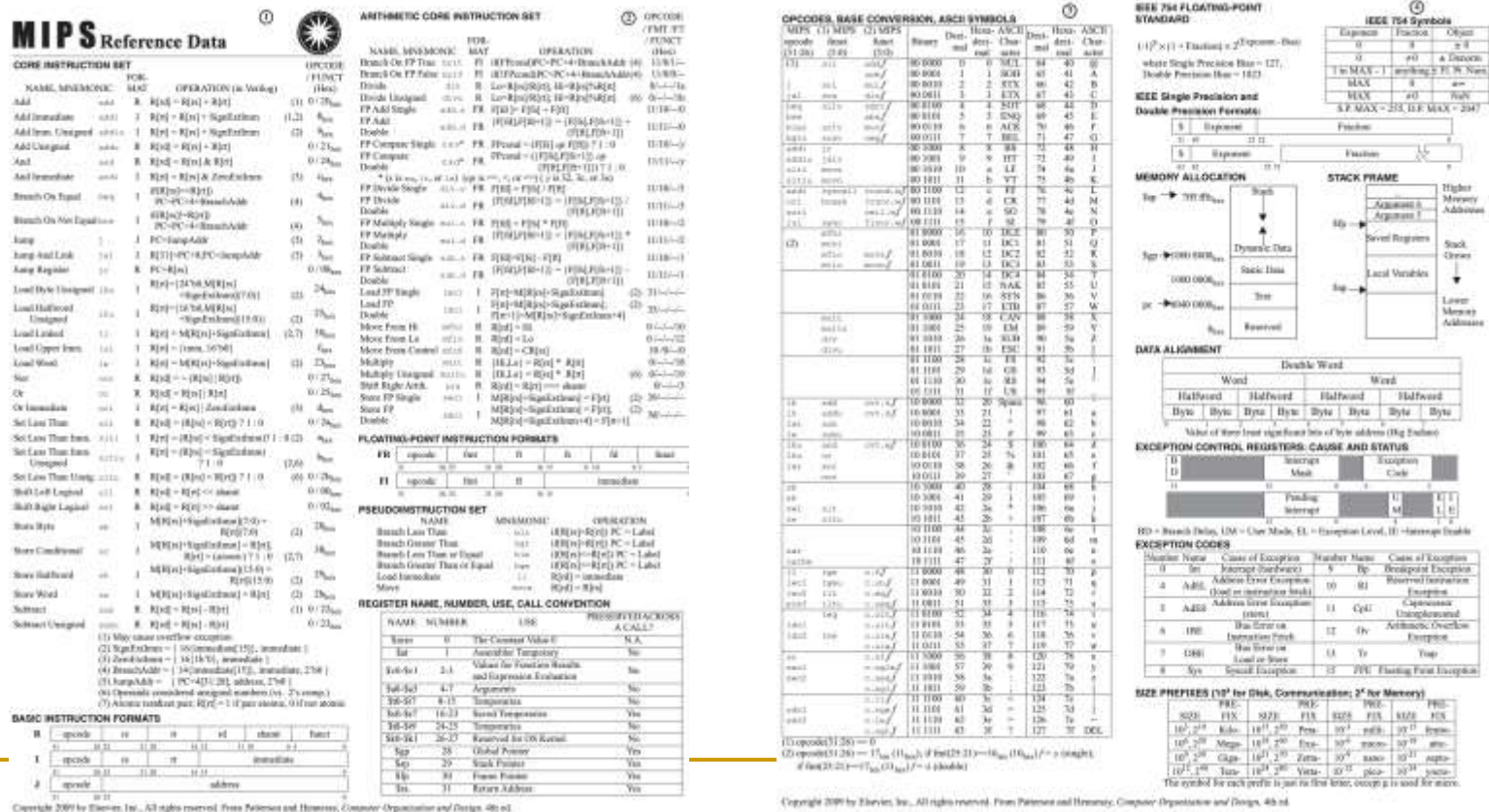
Semantics

if $\text{MEM}[\text{PC}] == \text{jalr rs}$
 $\$ra \leftarrow \text{PC} + 4$
 $\text{PC} \leftarrow \text{GPR}(\text{rs})$

[†] This is the incremented PC

- https://safari.ethz.ch/digitaltechnik/spring2023/lib/exe/fetch.php?media=mips_reference_data.pdf

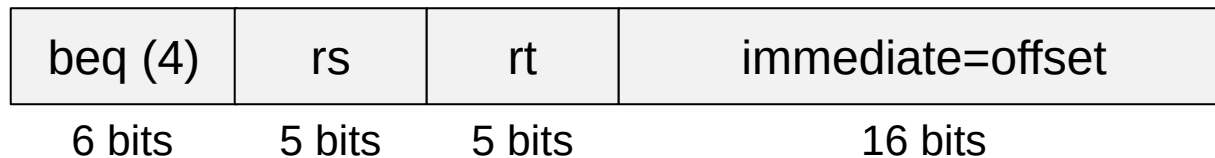
- On the course website



Conditional Branch Instructions

■ beq (Branch if Equal)

beq \$s0, \$s1, offset # \$s0=**rs**, \$s1=**rt**



I-Type

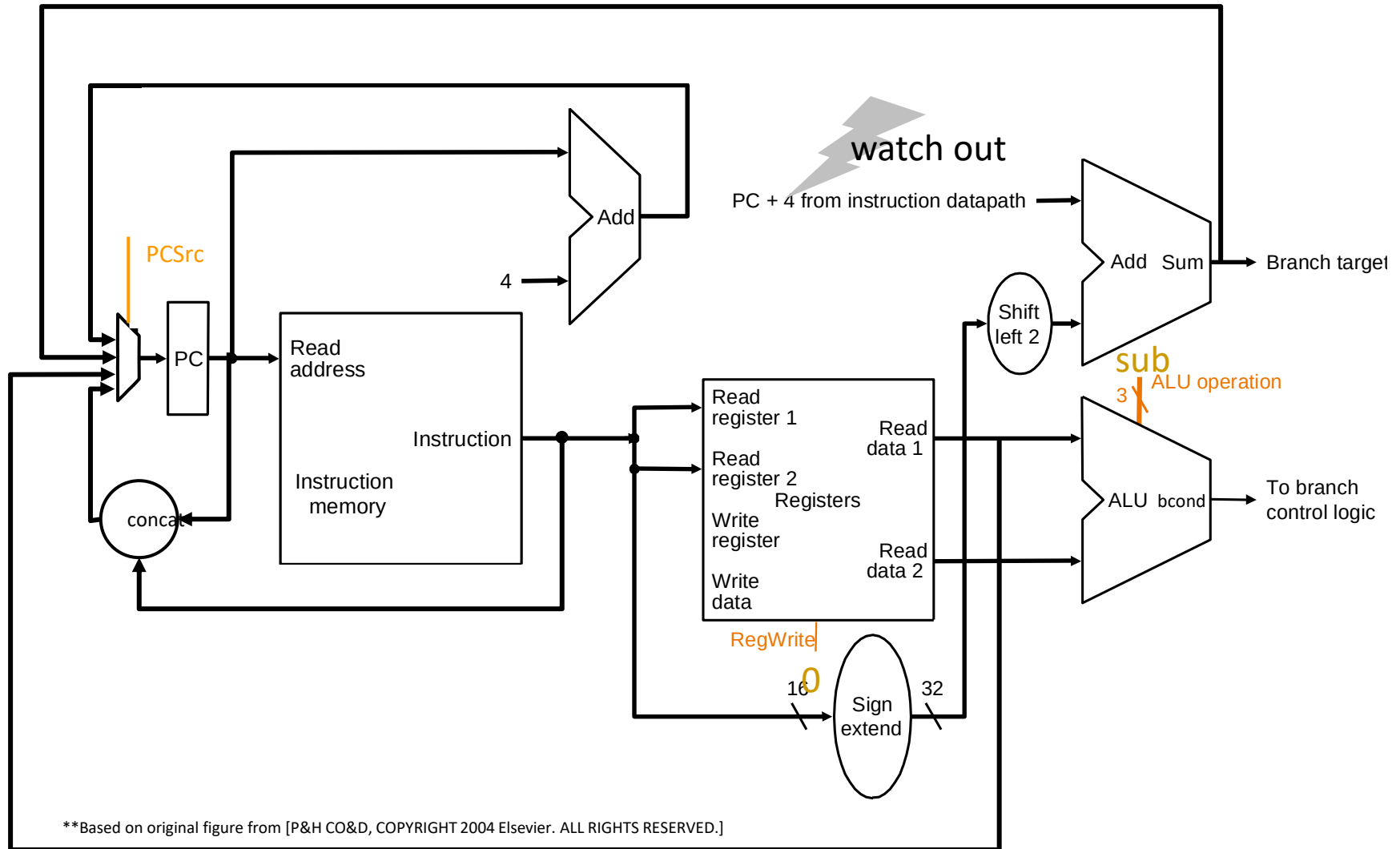
■ Semantics (assuming no branch delay slot)

if $\text{MEM}[\text{PC}] == \text{beq } rs \text{ } rt \text{ } \text{immediate}_{16}$
 $\text{target} = \text{PC}^{\dagger} + \text{sign-extend}(\text{immediate}) \times 4$
 if $\text{GPR}[rs] == \text{GPR}[rt]$ then $\text{PC} \leftarrow \text{target}$
 else $\text{PC} \leftarrow \text{PC} + 4$

□ Variations: beq, bne, blez, bgtz

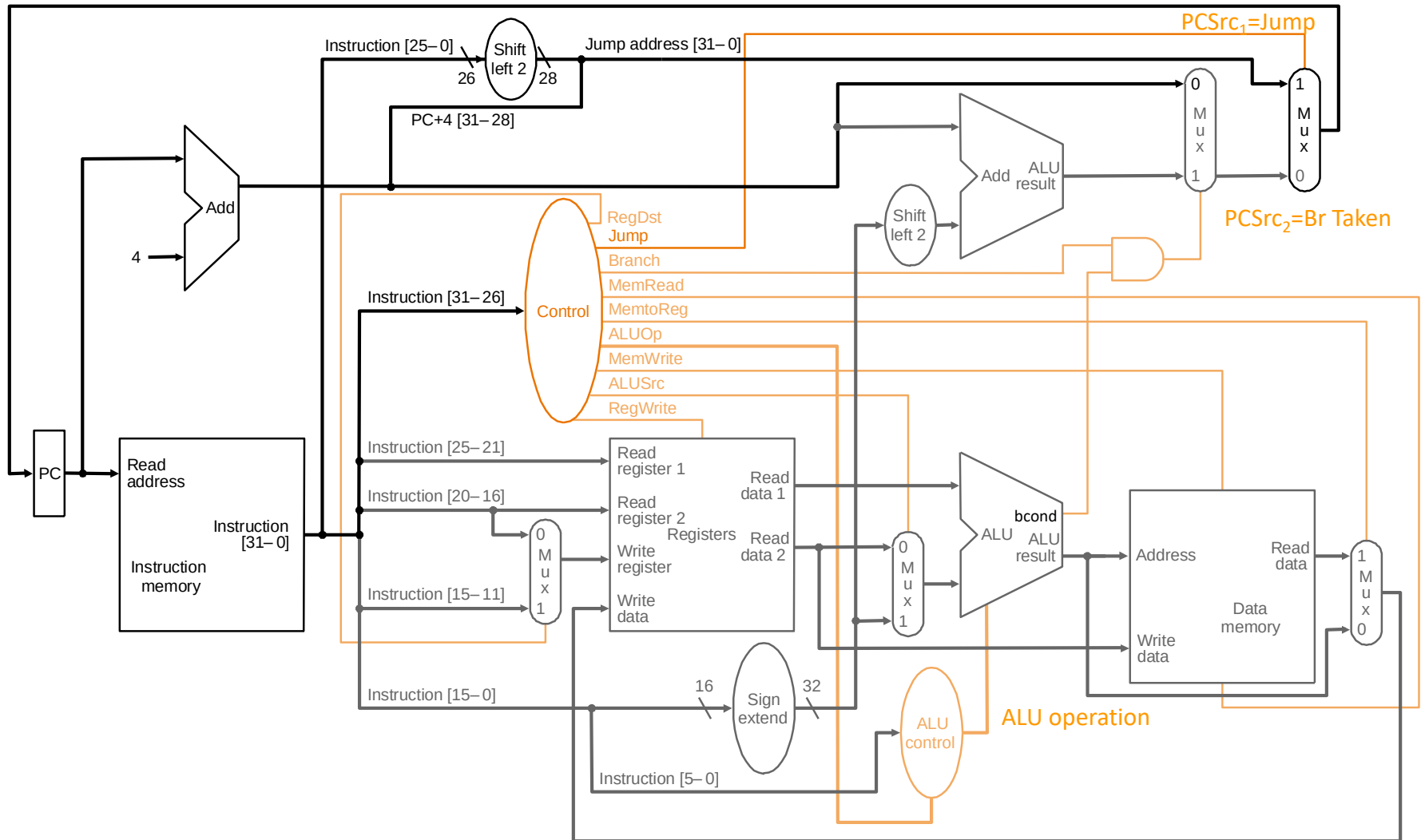
[†] This is the incremented PC

Conditional Branch Datapath (for you to finish)



Foreshadowing: How to uphold the delayed branch semantics?

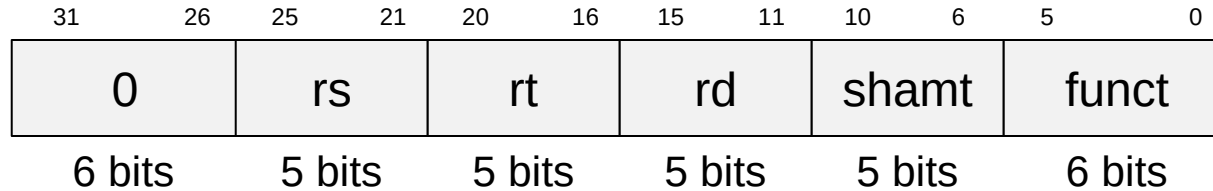
Putting It All Together



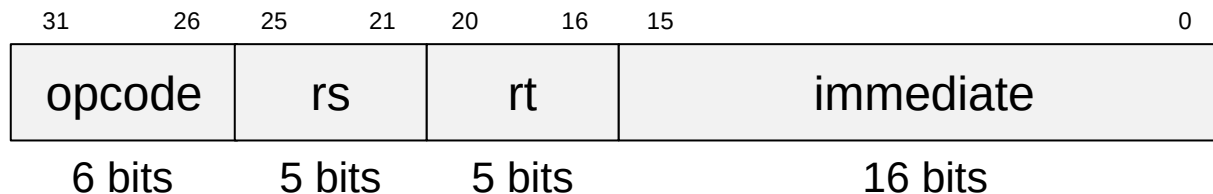
Single-Cycle Control Logic

Single-Cycle Hardwired Control

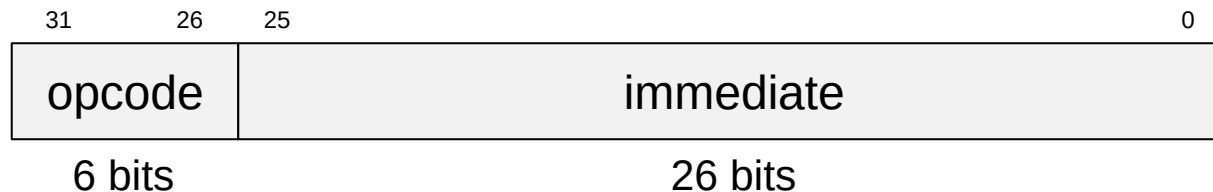
■ As combinational function of $\text{Inst} = \text{MEM}[\text{PC}]$



R-Type



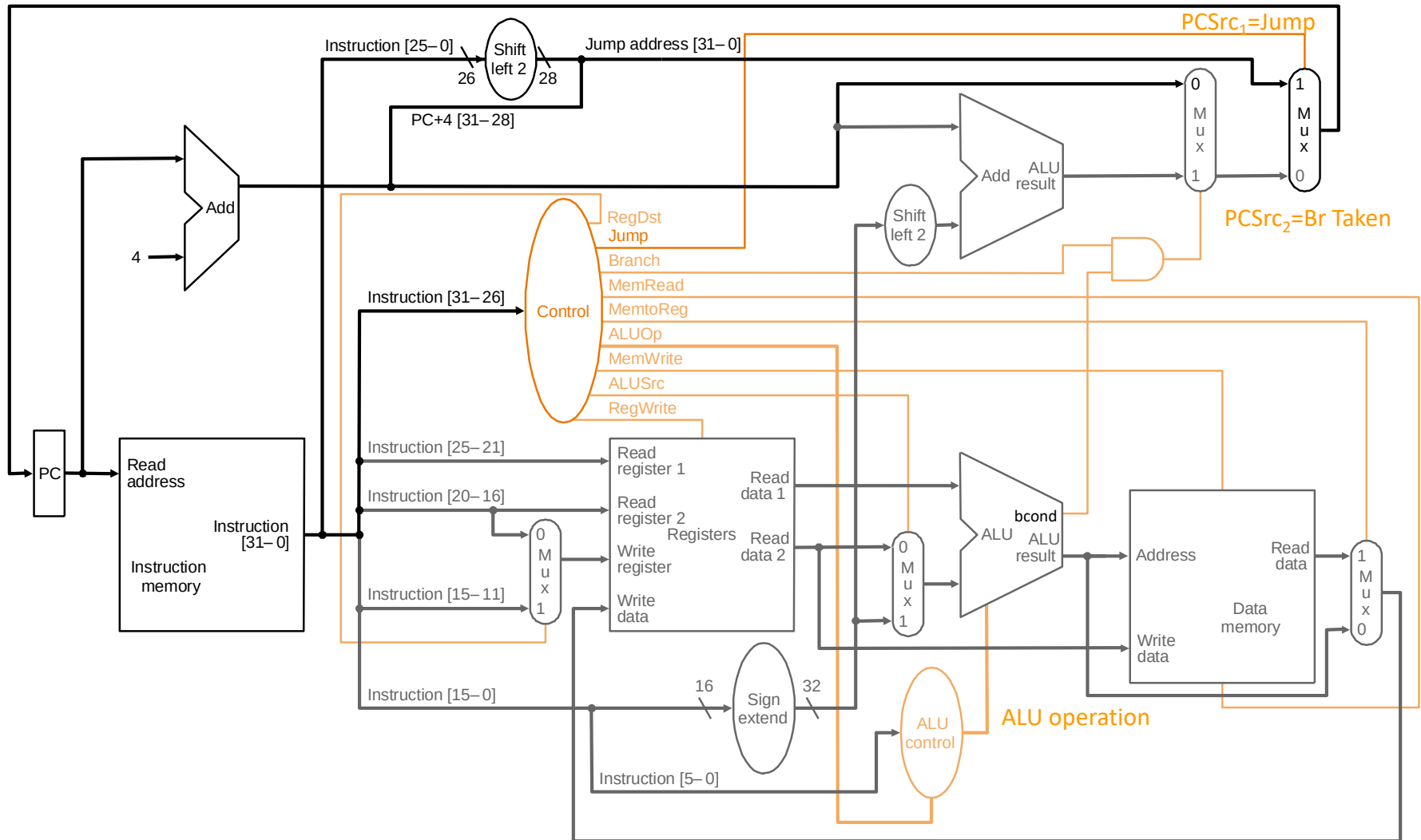
I-Type



J-Type

- Consider
 - ❑ All R-type and I-type ALU instructions
 - ❑ lw and sw
 - ❑ beq, bne, blez, bgtz
 - ❑ j, jr, jal, jalr omitted

Generate Control Signals (in Orange Color)



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JAL, JR, JALR omitted

Single-Bit Control Signals (I)

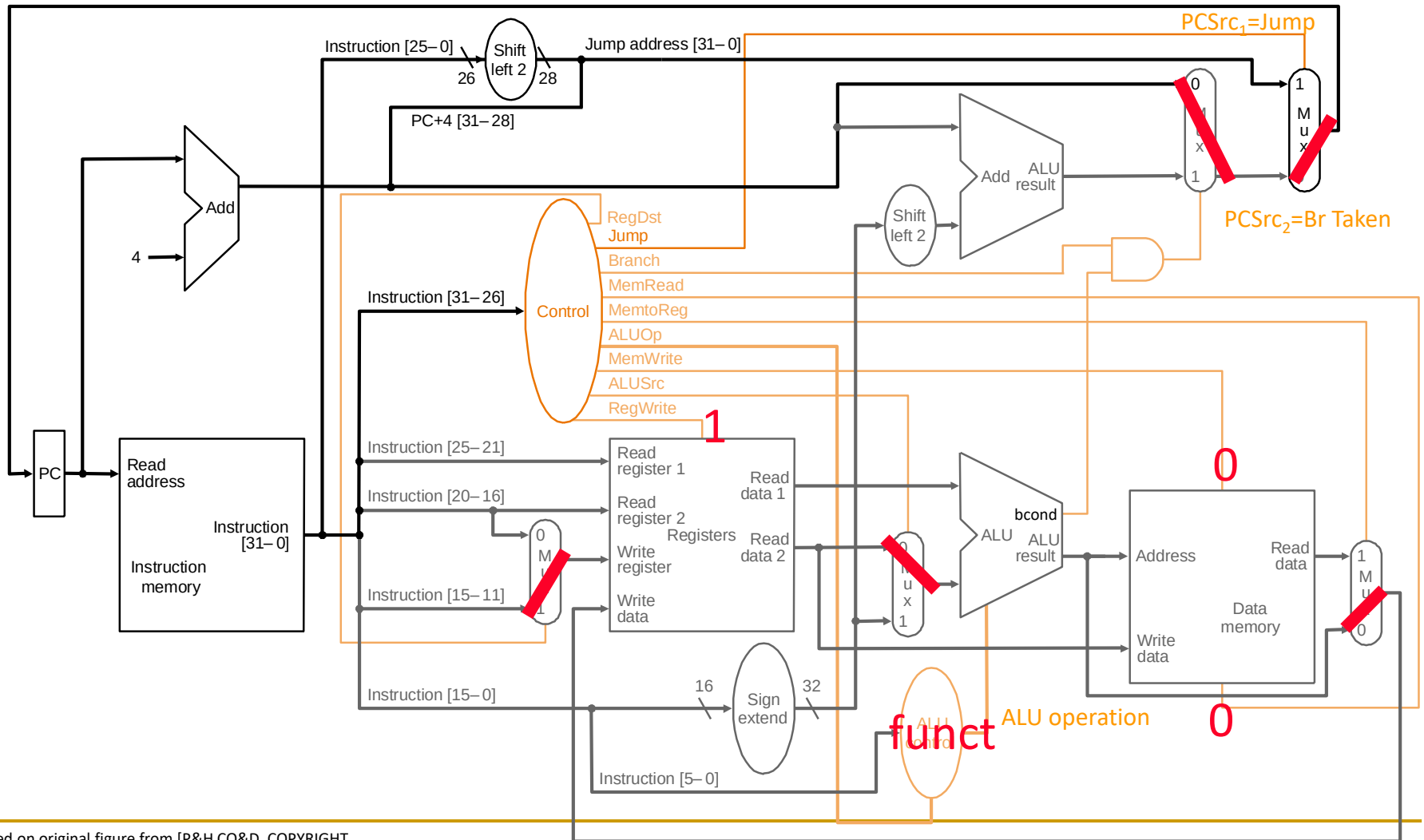
	When De-asserted	When asserted	Equation
RegDst	GPR write select according to rt , i.e., inst[20:16]	GPR write select according to rd , i.e., inst[15:11]	<code>opcode==0</code>
ALUSrc	2 nd ALU input from 2 nd GPR read port	2 nd ALU input from sign-extended 16-bit immediate	<code>(opcode!=0) &&</code> <code>(opcode!=BEQ) &&</code> <code>(opcode!=BNE)</code>
MemtoReg	Steer ALU result to GPR write port	Steer memory output to GPR write port	<code>opcode==LW</code>
RegWrite	GPR write disabled	GPR write enabled	<code>(opcode!=SW) &&</code> <code>(opcode!=Bxx) &&</code> <code>(opcode!=J) &&</code> <code>(opcode!=JR))</code>

JAL and JALR require additional RegDst and MemtoReg options

Single-Bit Control Signals (II)

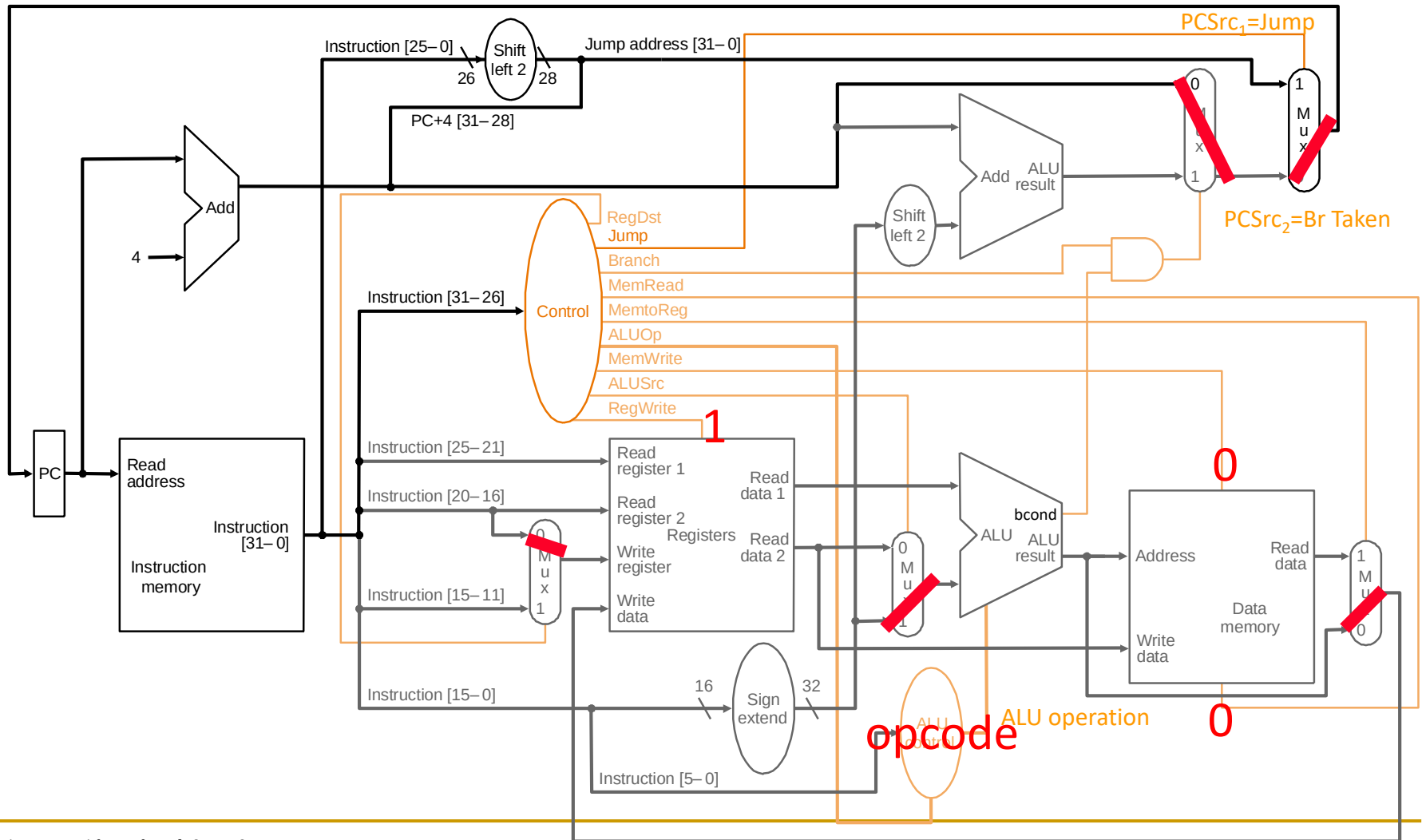
	When De-asserted	When asserted	Equation
MemRead	Memory read disabled	Memory read port returns load value	$\text{opcode} == \text{LW}$
MemWrite	Memory write disabled	Memory write enabled	$\text{opcode} == \text{SW}$
PCSrc ₁	According to PCSrc₂	next PC is based on 26-bit immediate jump target	$(\text{opcode} == \text{J}) \mid \mid$ $(\text{opcode} == \text{JAL})$
PCSrc ₂	next PC = PC + 4	next PC is based on 16-bit immediate branch target	$(\text{opcode} == \text{Bxx}) \ \&\&$ “bcond is satisfied”

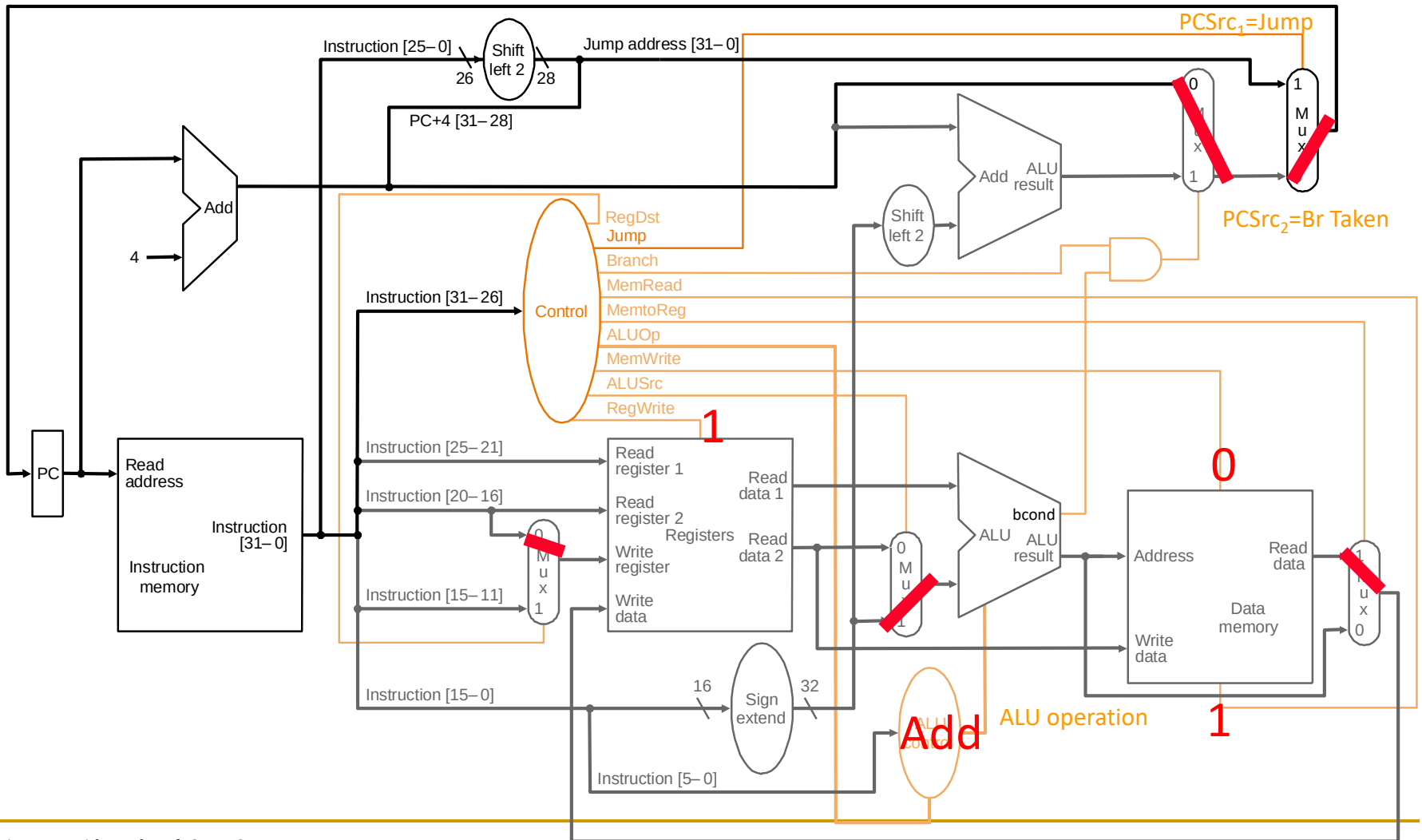
R-Type ALU

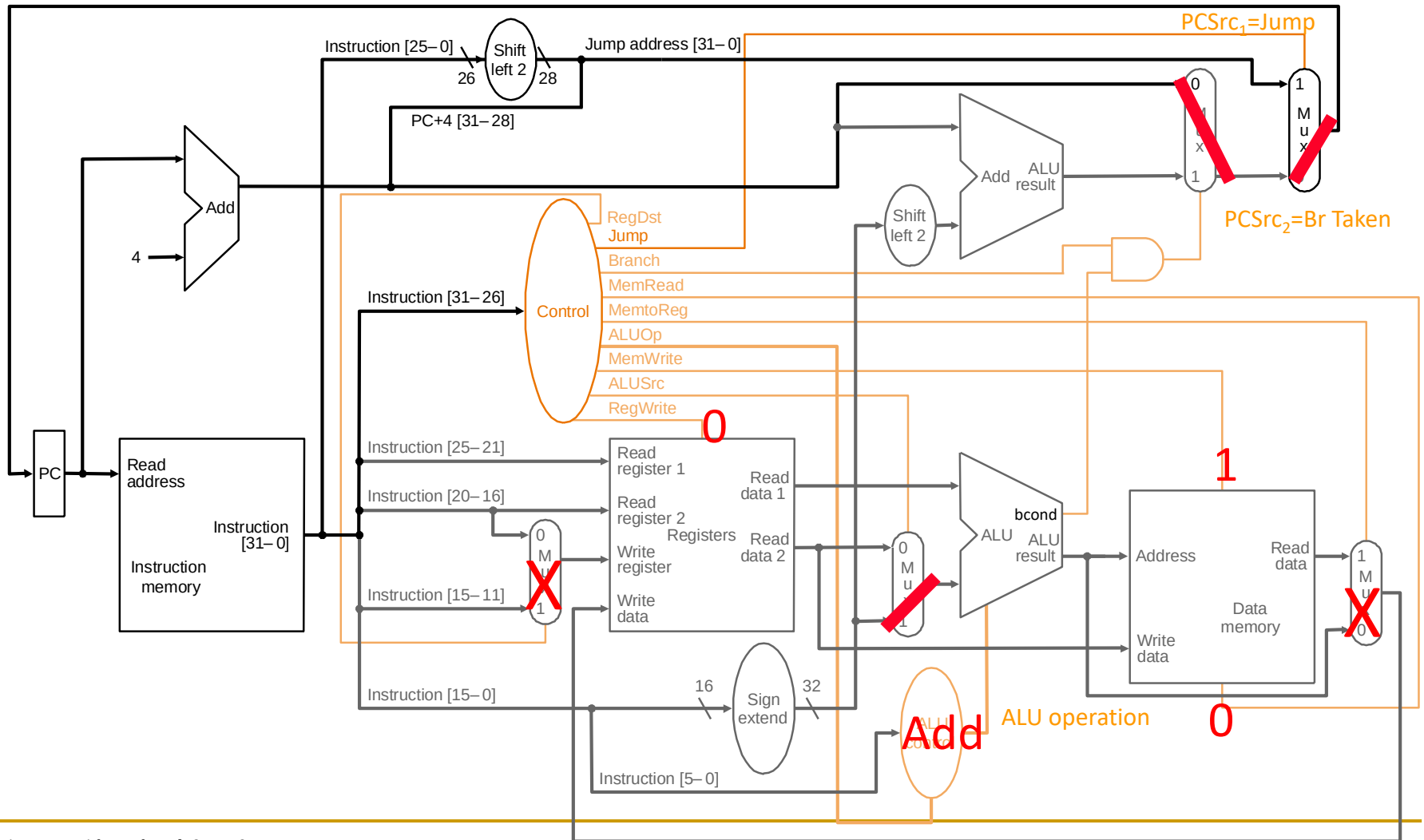


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I-Type ALU

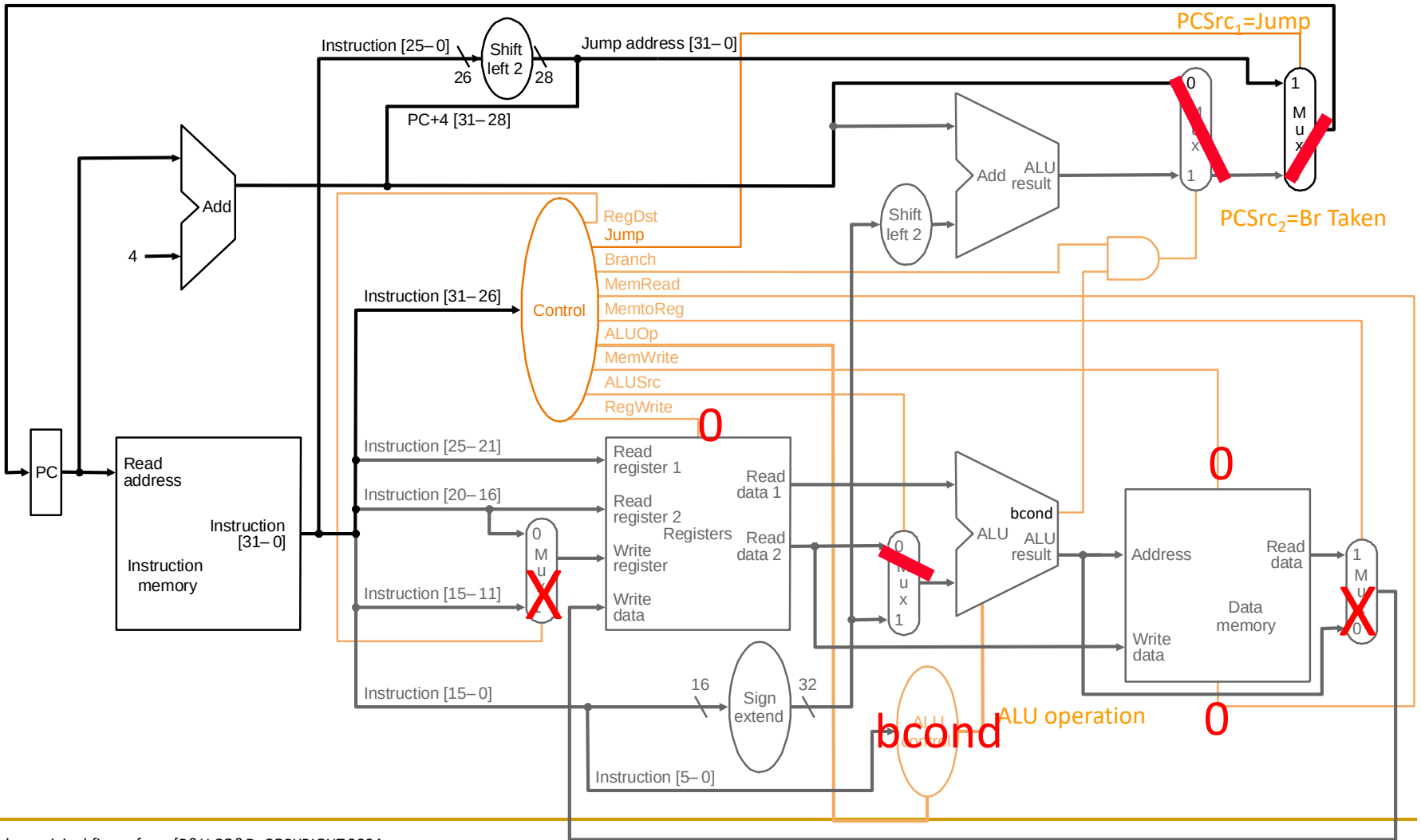






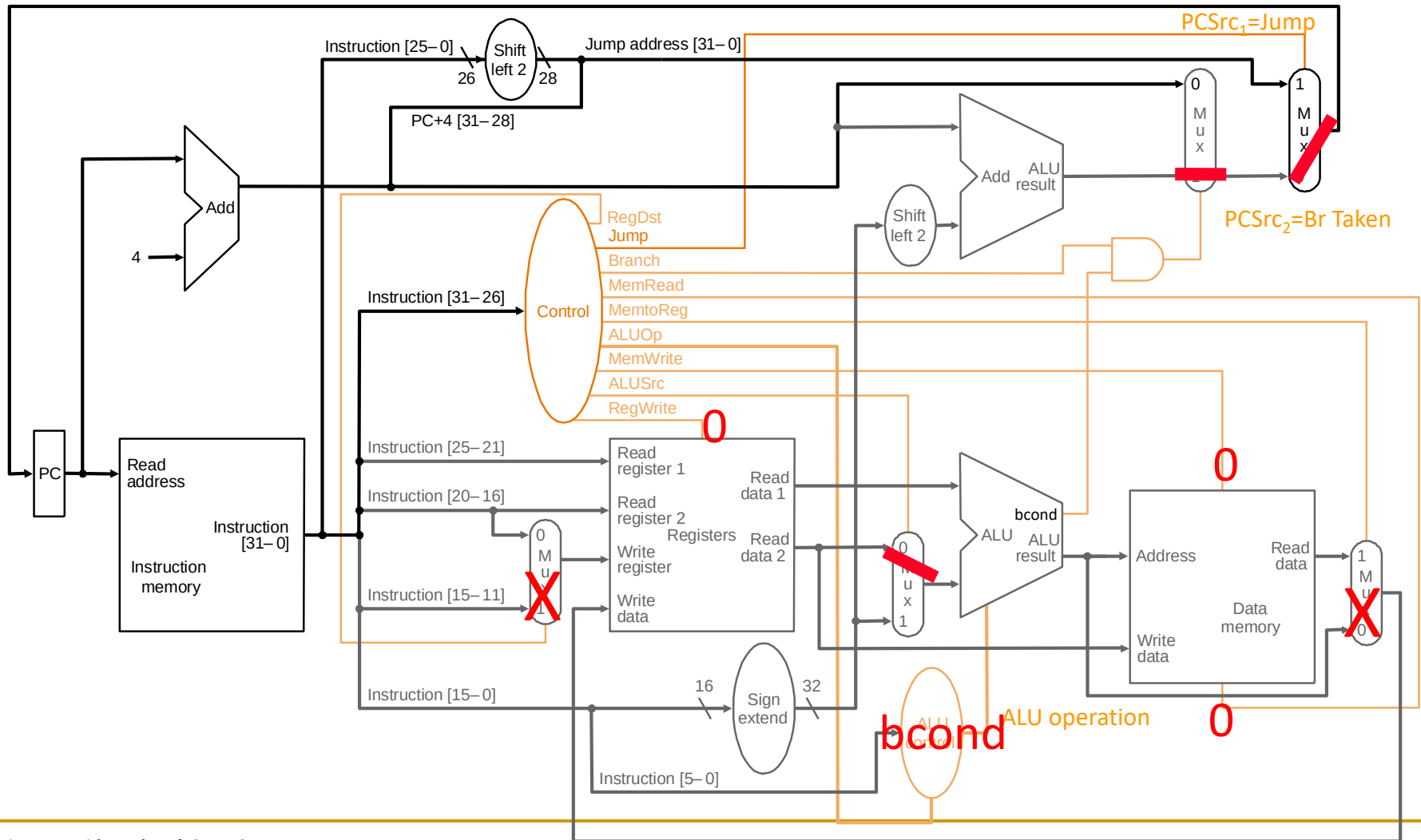
Branch (Not Taken)

Some control signals are dependent on the processing of data

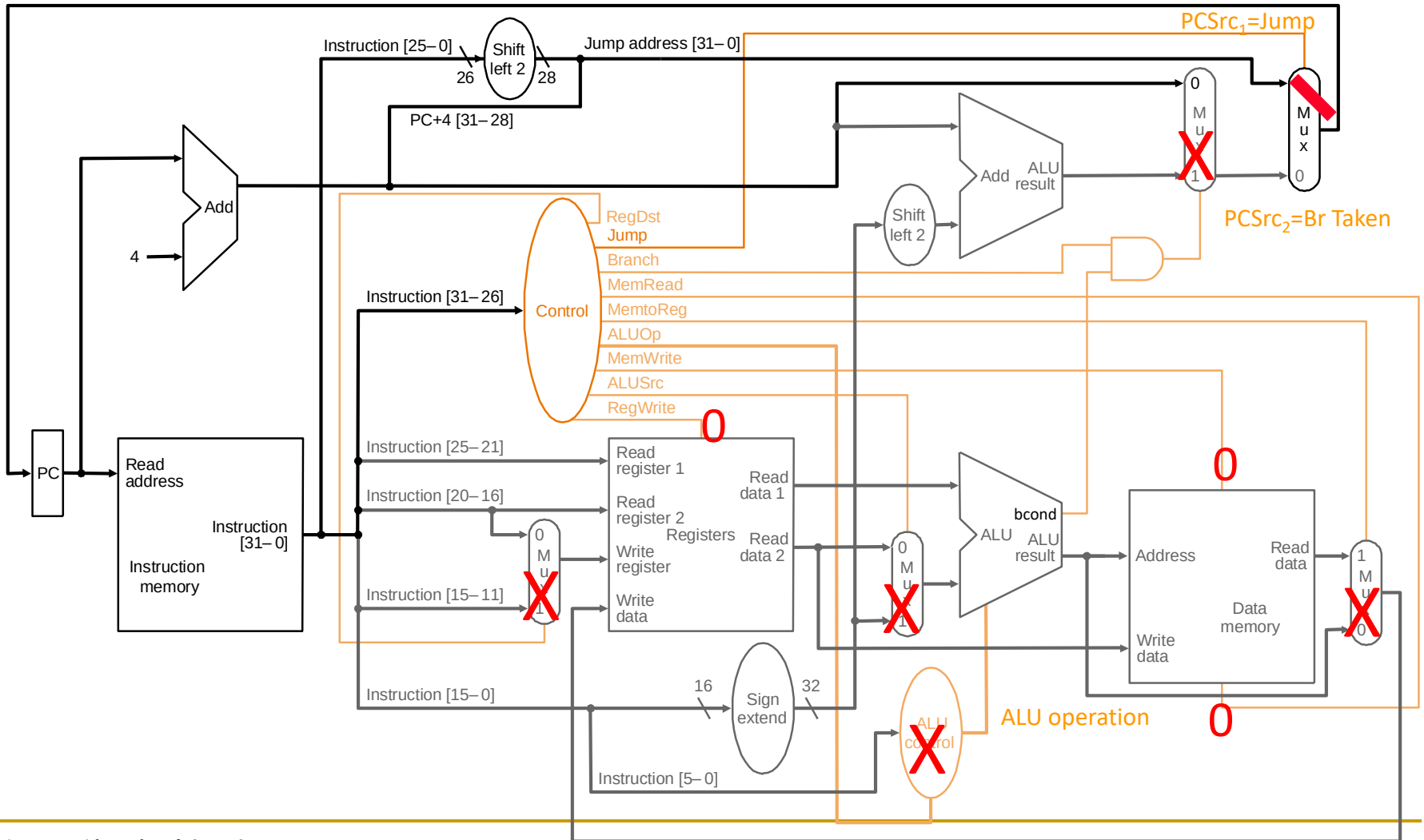


Branch (Taken)

Some control signals are dependent on the processing of data



Jump

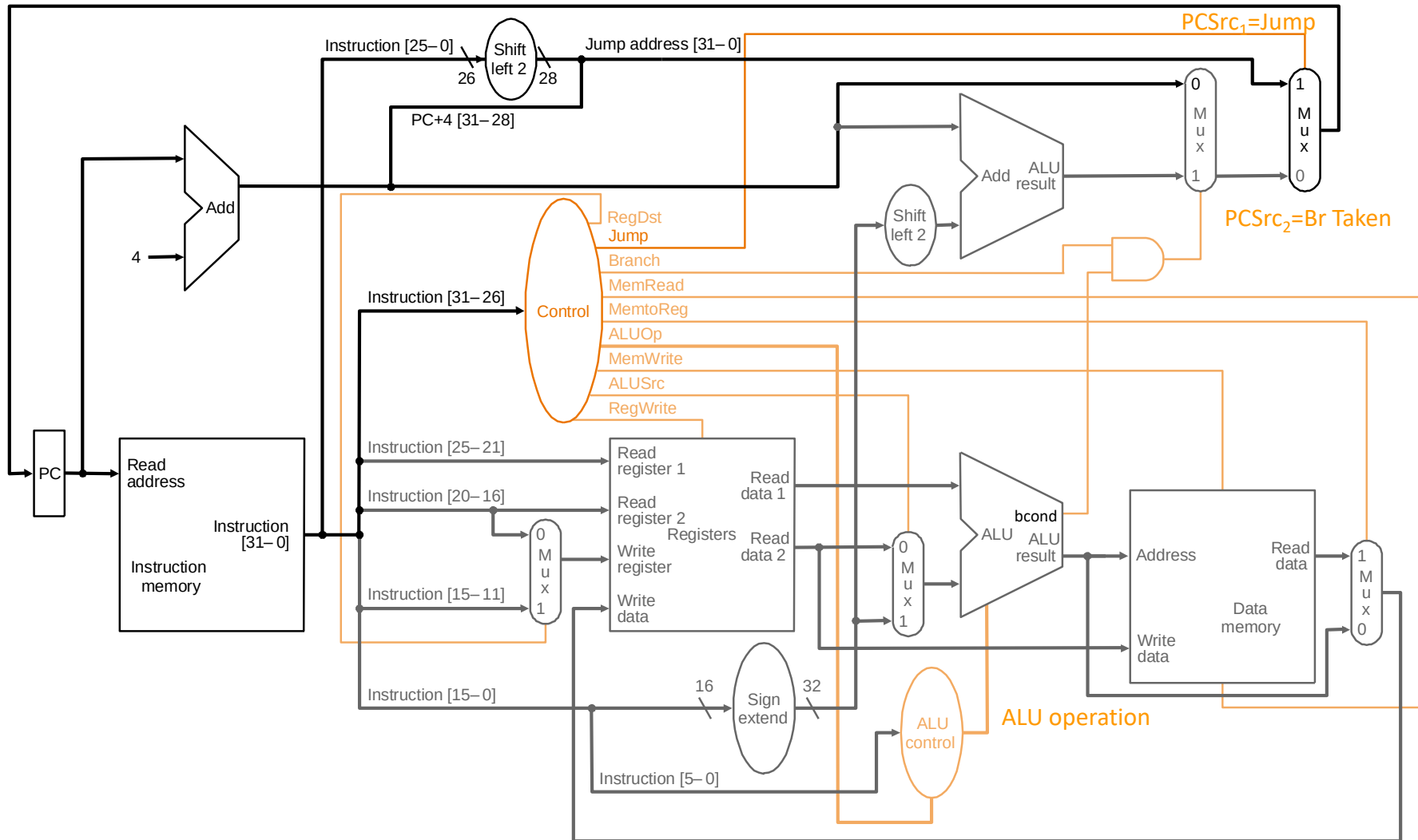


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What is in That Control Box?

- Combinational Logic → **Hardwired Control**
 - Idea: Control signals generated combinatorially based on bits in instruction encoding
- Sequential Logic → **Sequential Control**
 - Idea: A memory structure contains the control signals associated with an instruction
 - Called **Control Store**
- **Both types of control structure can be used in single-cycle processors**
 - Choice depends on latency of each structure + how much on the critical path control signal generation is, etc.

Review: Complete Single-Cycle Processor



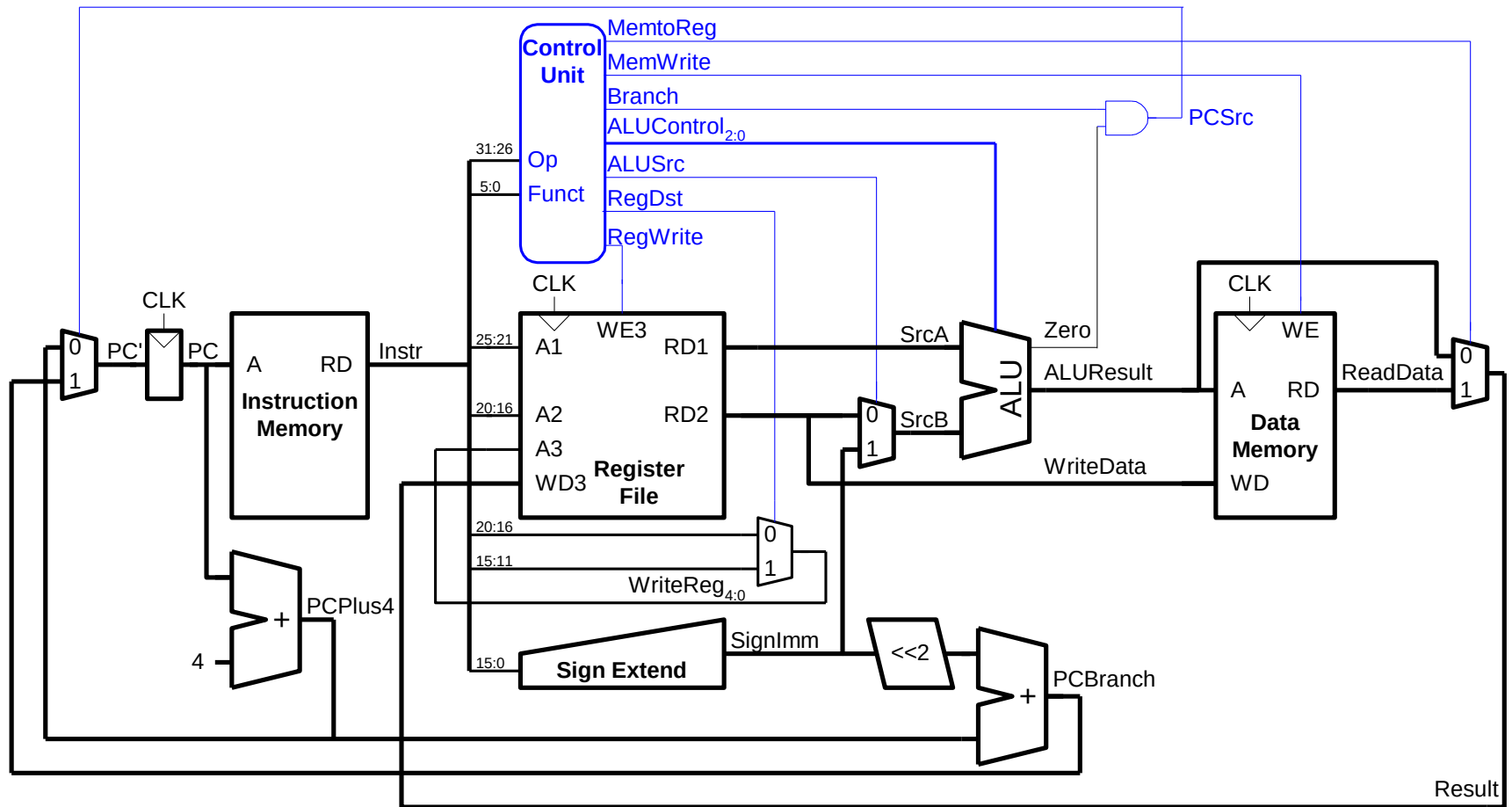
Another Single-Cycle MIPS Processor (from H&H)

See backup slides to reinforce the concepts we have covered.

They are to complement your reading:

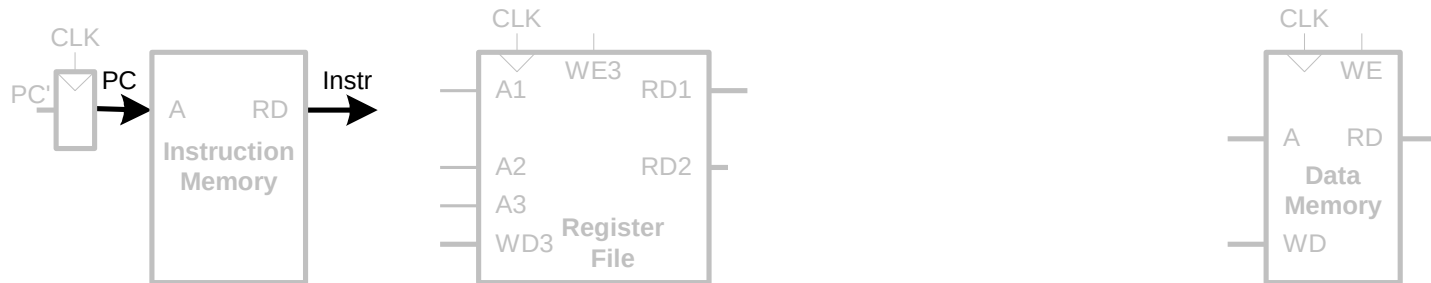
H&H, Chapter 7.1-7.3, 7.6

Another Complete Single-Cycle Processor



Example: Single-Cycle Datapath: lw fetch

■ **STEP 1:** Fetch instruction



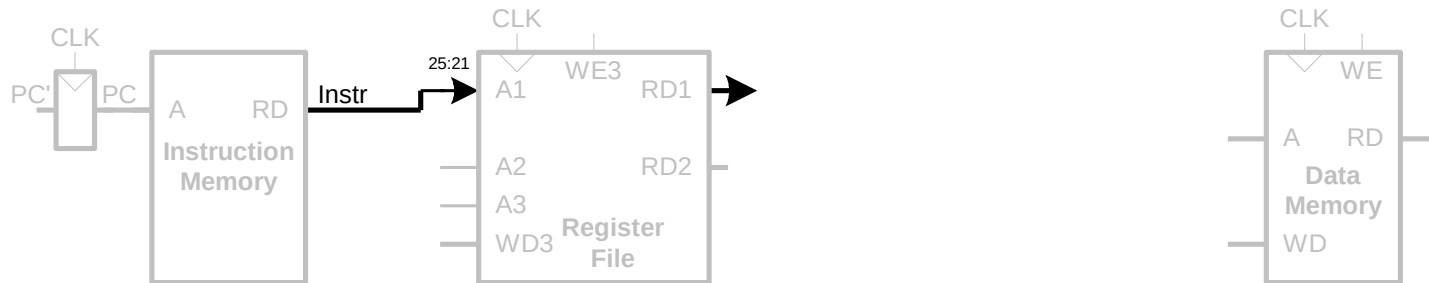
`lw $s3, 1($0)` # read memory word 1 into \$s3

I-Type



Single-Cycle Datapath: lw register read

■ **STEP 2:** Read source operands from register file



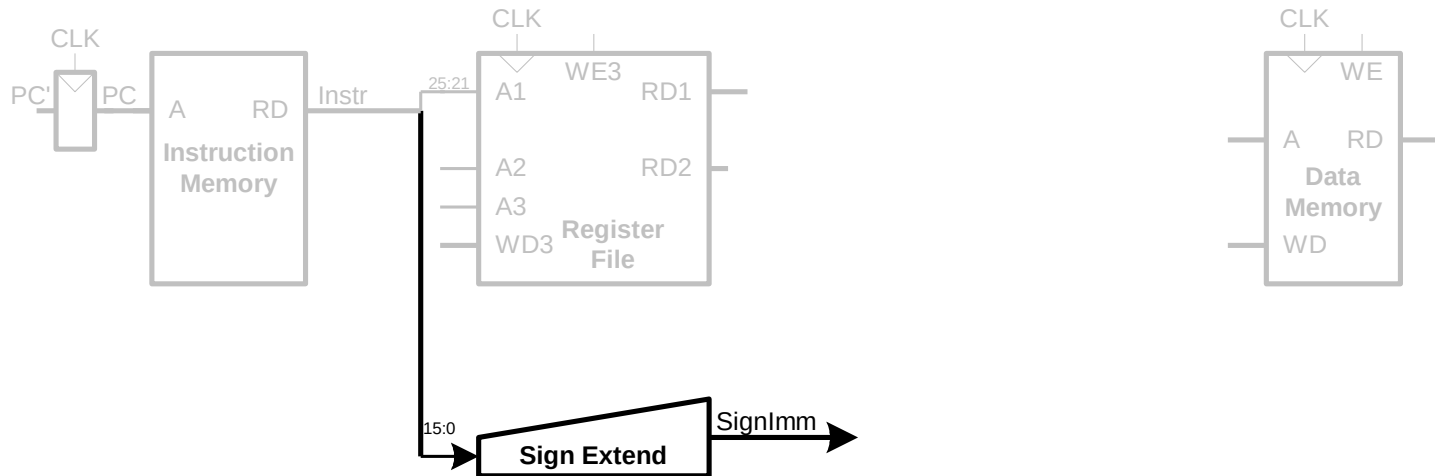
`lw $s3, 1($0)` # read memory word 1 into \$s3

I-Type

op	rs	rt	imm
6 bits	5 bits	5 bits	16 bits

Single-Cycle Datapath: lw immediate

■ **STEP 3:** Sign-extend the immediate



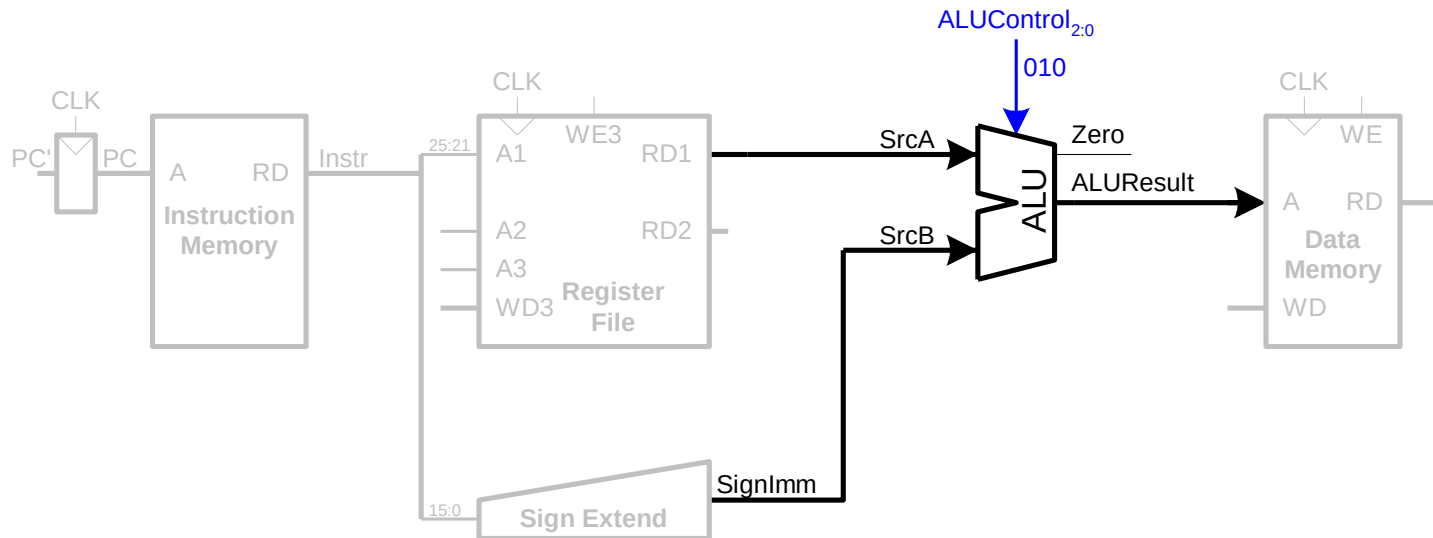
`lw $s3, 1($0)` # read memory word 1 into \$s3

I-Type

op	rs	rt	imm
6 bits	5 bits	5 bits	16 bits

Single-Cycle Datapath: lw address

■ **STEP 4:** Compute the memory address



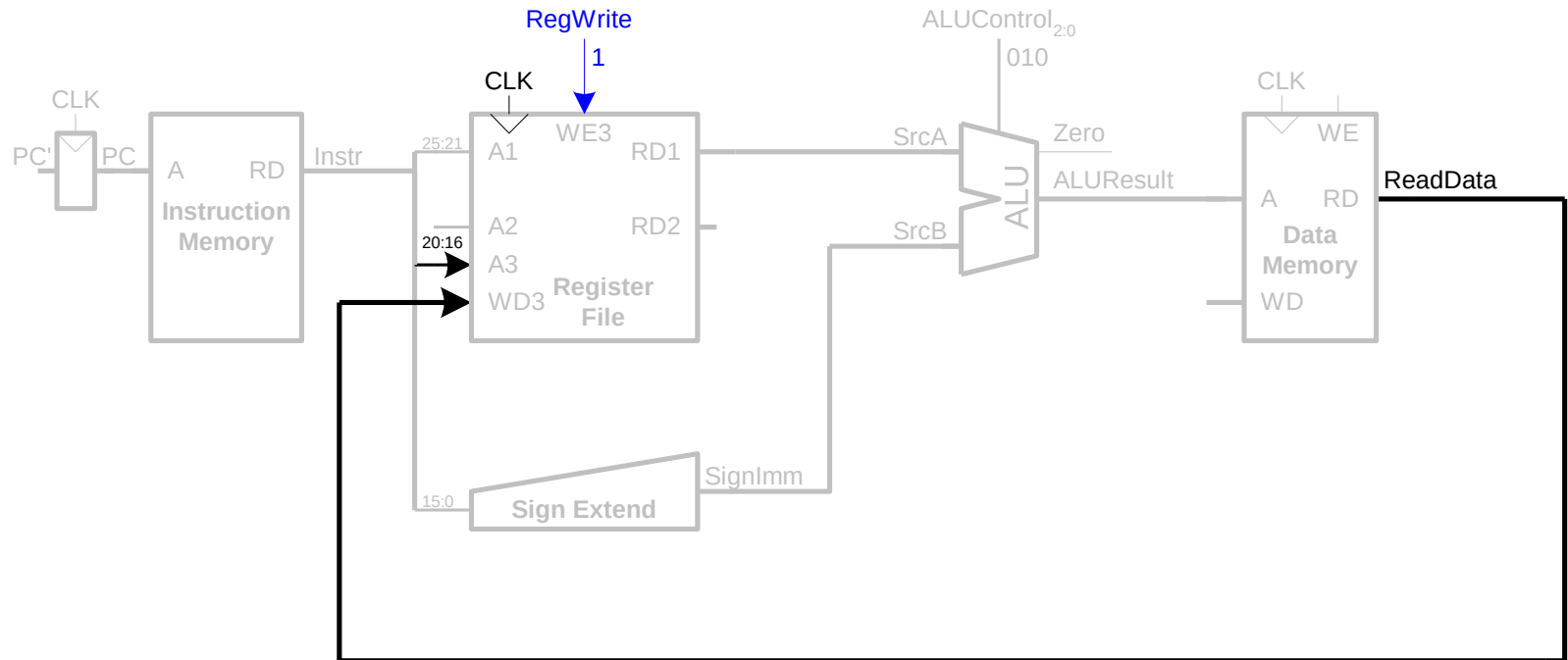
`lw $s3, 1($0)` # read memory word 1 into \$s3

I-Type

op	rs	rt	imm
6 bits	5 bits	5 bits	16 bits

Single-Cycle Datapath: lw memory read

■ **STEP 5:** Read from memory and write back to register file



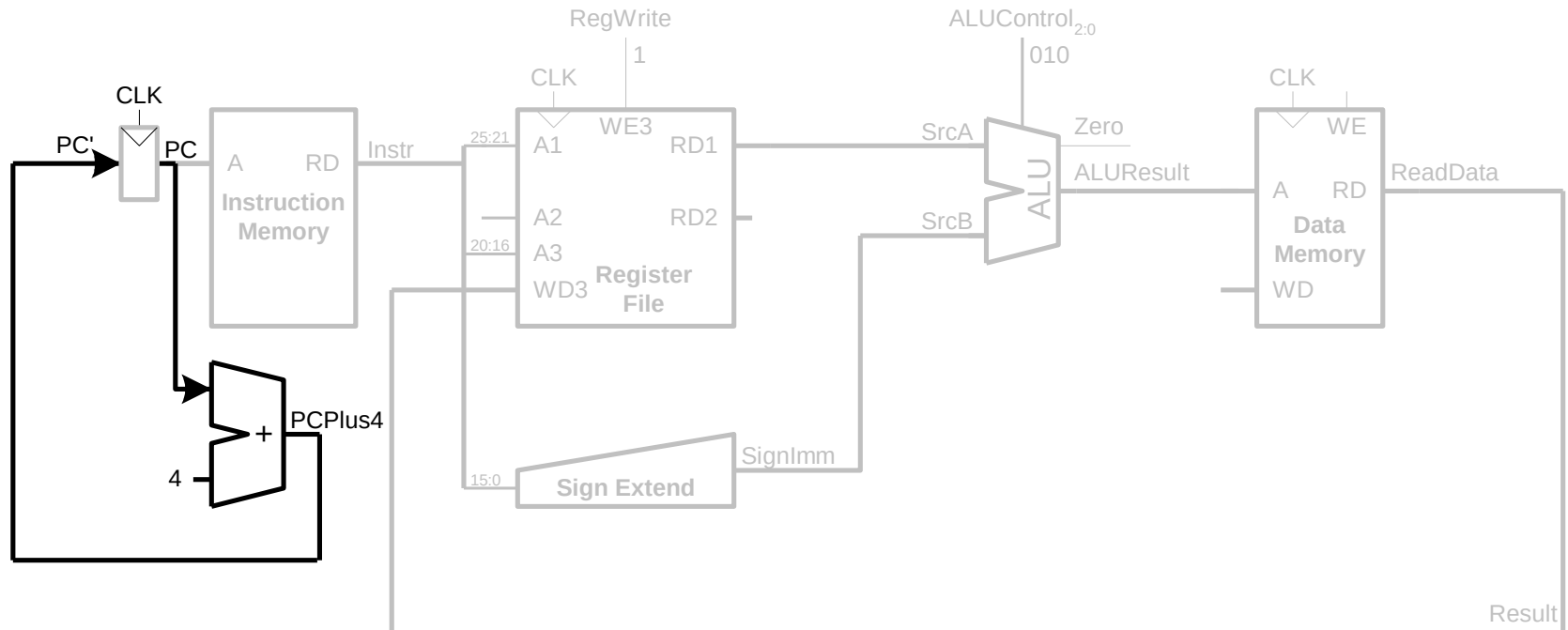
`lw $s3, 1($0)` # read memory word 1 into \$s3

I-Type

op	rs	rt	imm
6 bits	5 bits	5 bits	16 bits

Single-Cycle Datapath: lw PC increment

■ **STEP 6:** Determine address of next instruction



`lw $s3, 1($0)` # read memory word 1 into \$s3

I-Type

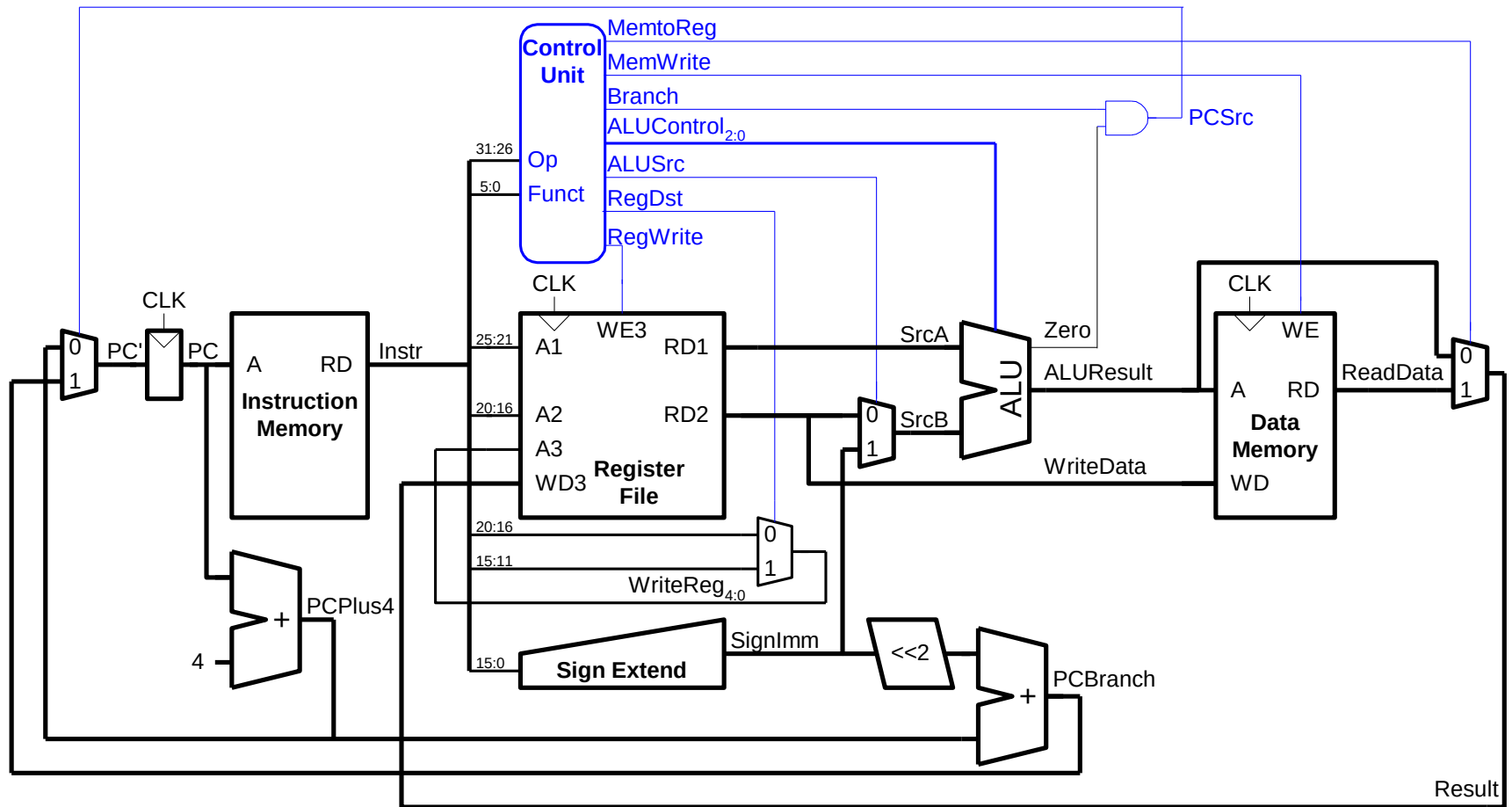
op	rs	rt	imm
6 bits	5 bits	5 bits	16 bits

Similarly, We Need to Design the Control Unit

- **Control signals** are generated by the decoder in control unit

Instruction	Op _{5:0}	RegWrite	RegDst	AluSrc	Branch	MemWrite	MemtoReg	ALUOp _{1:0}	Jump
R-type	000000	1	1	0	0	0	0	10	0
lw	100011	1	0	1	0	0	1	00	0
sw	101011	0	X	1	0	1	X	00	0
beq	000100	0	X	0	1	0	X	01	0
addi	001000	1	0	1	0	0	0	00	0
j	000010	0	X	X	X	0	X	XX	1

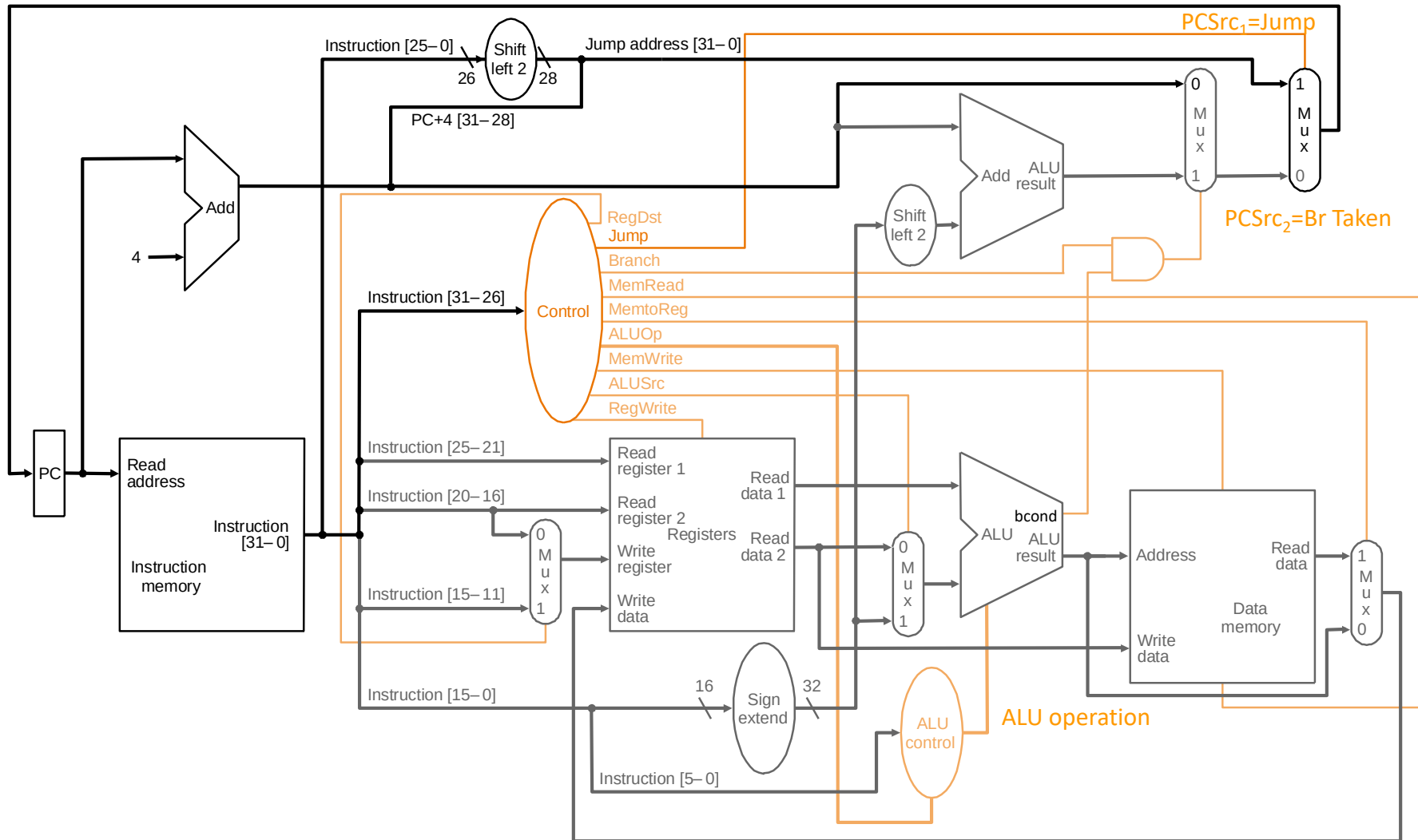
Another Complete Single-Cycle Processor (H&H)



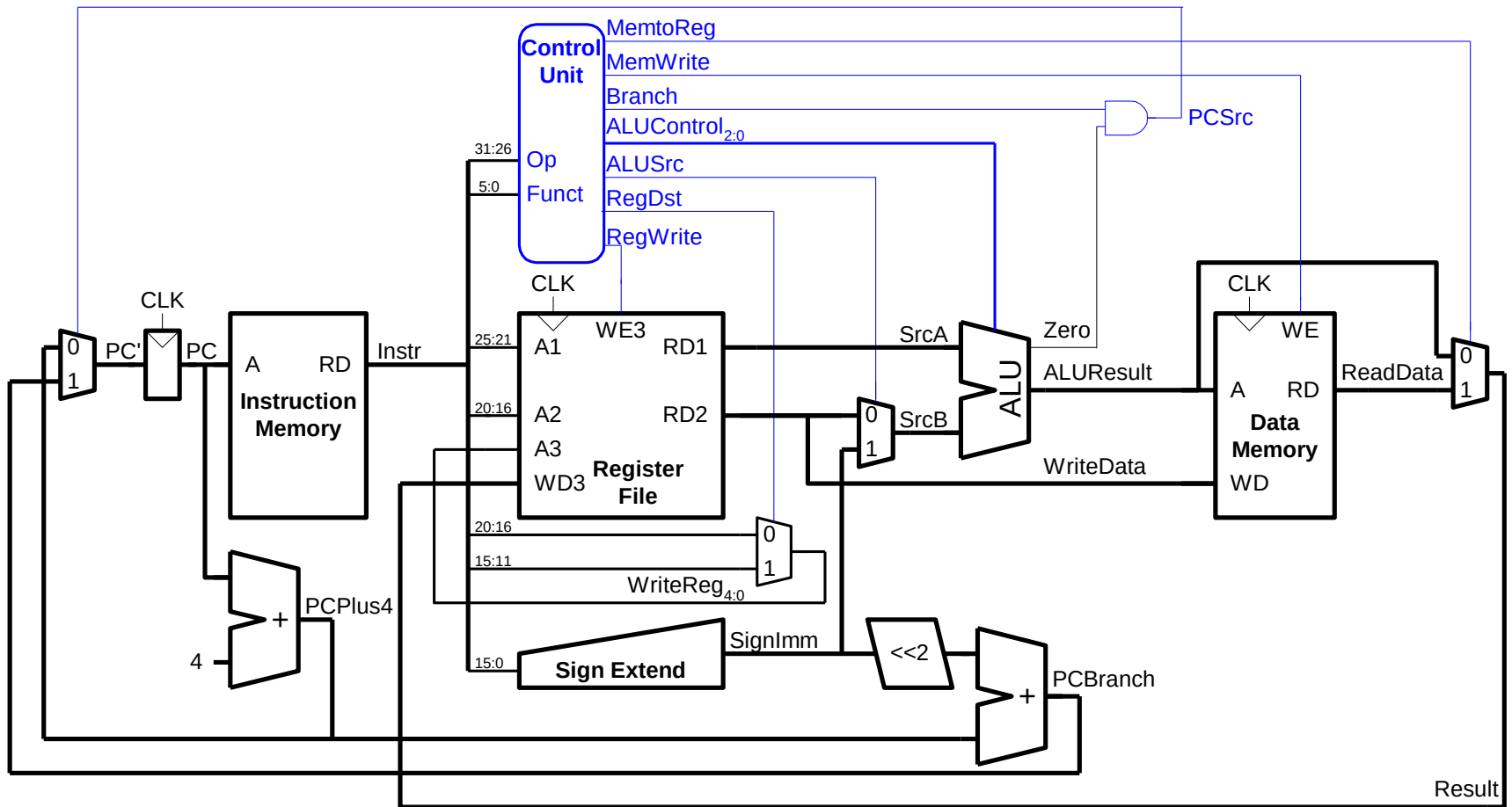
Your Reading Assignment

- Please go over and internalize the **Lecture Slides & Backup Slides**
- Please do your readings from the H&H Book
 - H&H, Chapter 7.1-7.3, 7.6

Single-Cycle Uarch I (We Developed in Lectures)



Single-Cycle Uarch II (In Your Readings)



Evaluating the Single-Cycle Microarchitecture

A Single-Cycle Microarchitecture

- Is *this* a good idea/design?
- When is this a good design?
- When is this a bad design?
- How can we design a better microarchitecture?

Performance Analysis Basics

Recall: Performance Analysis Basics

- Execution time of a single instruction
 - **{CPI} x {clock cycle time}**
 - CPI: Number of cycles it takes to execute an instruction
- Execution time of an entire program
 - Sum over all instructions [**{CPI} x {clock cycle time}**]
 - **{# of instructions} x {Average CPI} x {clock cycle time}**

Processor Performance

- **How fast is my program?**
 - Every program consists of a series of instructions
 - Each instruction needs to be executed

Processor Performance

■ How fast is my program?

- Every program consists of a series of instructions
- Each instruction needs to be executed

■ How fast are my instructions?

- Instructions are realized on the hardware
- Each instruction can take one or more clock cycles to complete
- *Cycles per Instruction = CPI*

Processor Performance

■ How fast is my program?

- Every program consists of a series of instructions
- Each instruction needs to be executed

■ How fast are my instructions?

- Instructions are realized on the hardware
- Each instruction can take one or more clock cycles to complete
- *Cycles per Instruction = CPI*

■ How long is one clock cycle?

- The critical path determines how much time one cycle requires = *clock period*
- $1/\text{clock period} = \text{clock frequency}$ = how many clock cycles are in each second

Processor Performance

■ As a general formula

- Our program consists of executing **N** instructions
- Our processor needs **CPI** cycles (on average) for each instruction
- The clock frequency of the processor is **f**
 - ➔ the clock period is therefore **T**=1/f

Processor Performance

■ As a general formula

- Our program consists of executing **N** instructions
- Our processor needs **CPI** cycles (on average) for each instruction
- The clock frequency of the processor is **f**
 - the clock period is therefore **T=1/f**

■ Our program executes in

$$N \times CPI \times (1/f) =$$

$$N \times CPI \times T \text{ seconds}$$

Performance Analysis of Our Single-Cycle Design

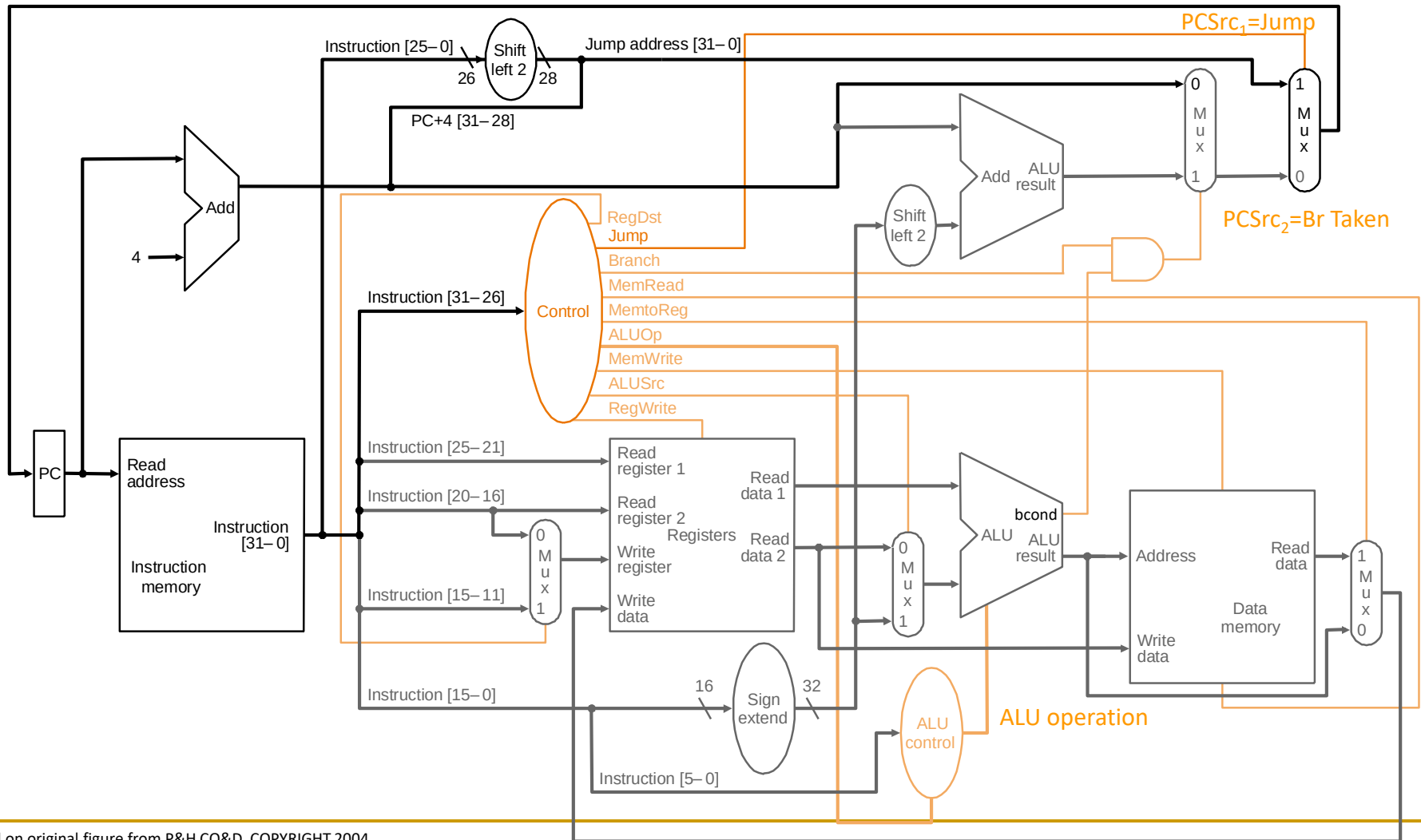
A Single-Cycle Microarchitecture: Analysis

- Every instruction takes 1 cycle to execute
 - CPI (Cycles per instruction) is strictly 1
- How long each instruction takes is determined by how long the slowest instruction takes to execute
 - Even though many instructions do not need that long to execute
- Clock cycle time of the microarchitecture is determined by how long it takes to complete the slowest instruction
 - Critical path of the design is determined by the processing time of the slowest instruction

What is the Slowest Instruction to Process?

- Let's go back to the basics
- All six phases of the instruction processing cycle take a *single machine clock cycle* to complete
 - Fetch
 - 1. Instruction fetch (IF)
 - Decode
 - 2. Instruction decode and
 - Evaluate Address
 - register operand fetch (ID/RF)
 - Fetch Operands
 - 3. Execute/Evaluate memory address (EX/AG)
 - Execute
 - 4. Memory operand fetch (MEM)
 - Store Result
 - 5. Store/writeback result (WB)
- Does every instruction take the same time (latency) to complete?

Let's Find the Critical Path

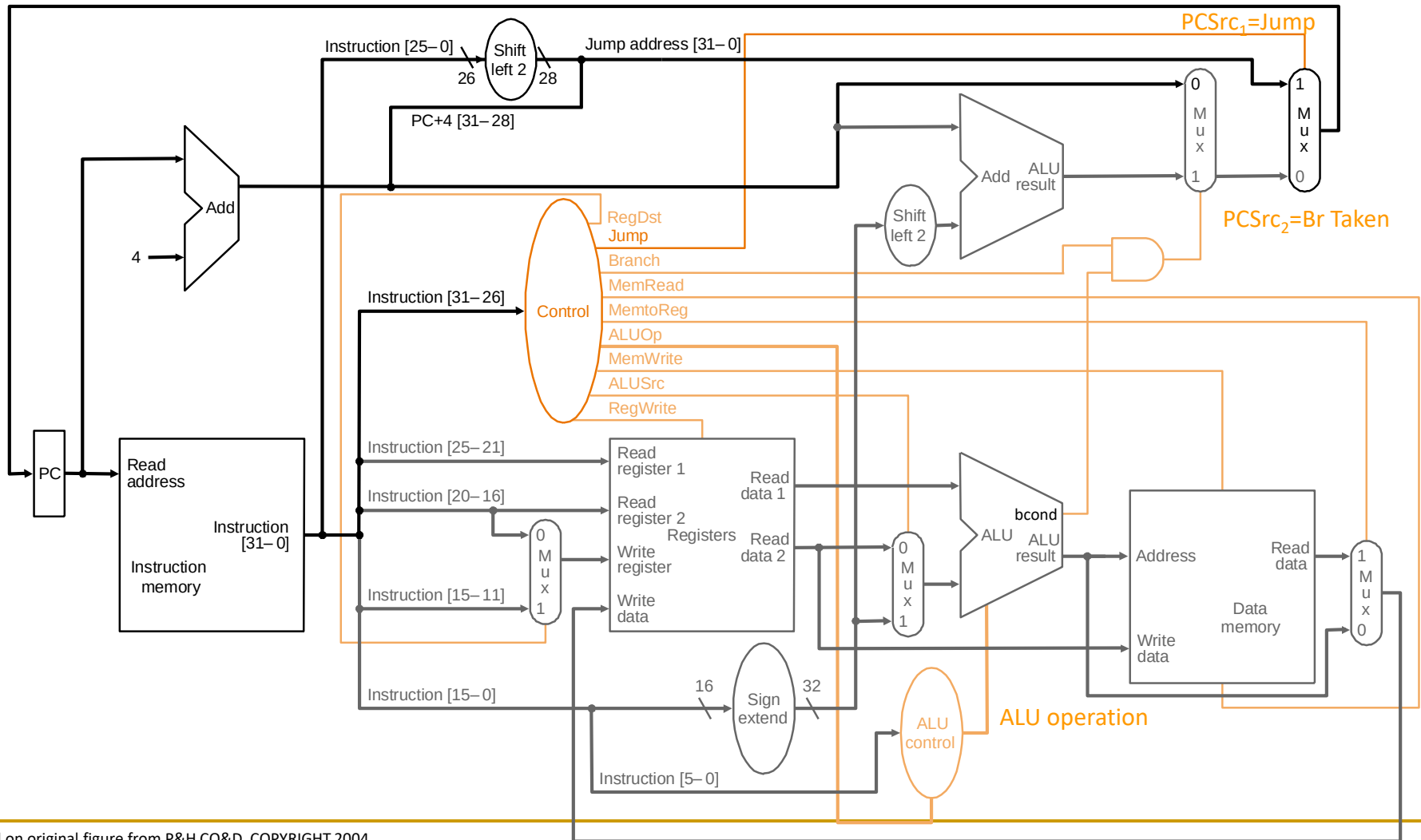


Example Single-Cycle Datapath Analysis

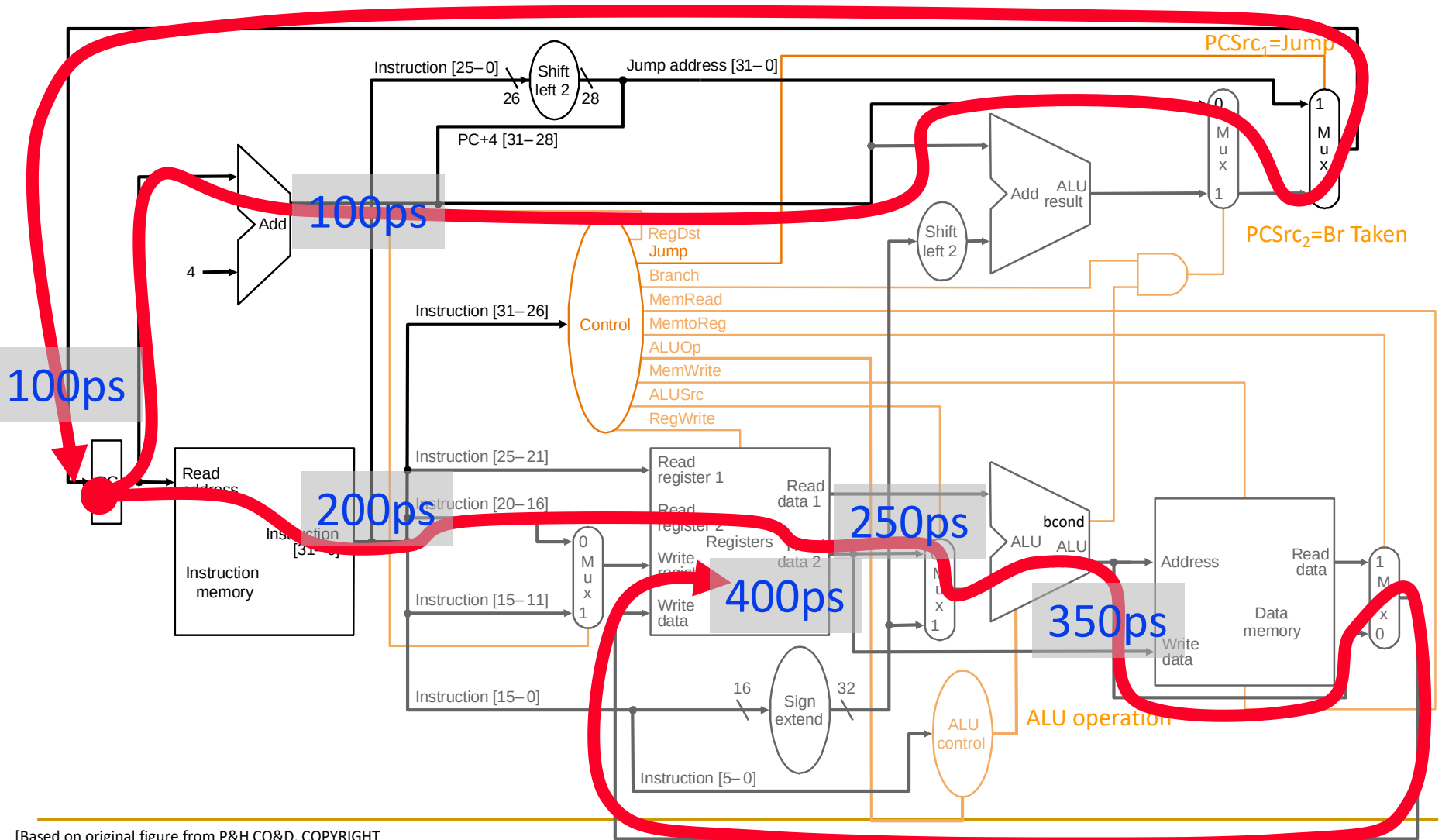
- Assume (for the design in the previous slide)
 - ❑ memory units (read or write): 200 ps
 - ❑ ALU and adders: 100 ps
 - ❑ register file (read or write): 50 ps
 - ❑ other logic or wire delay: 0 ps

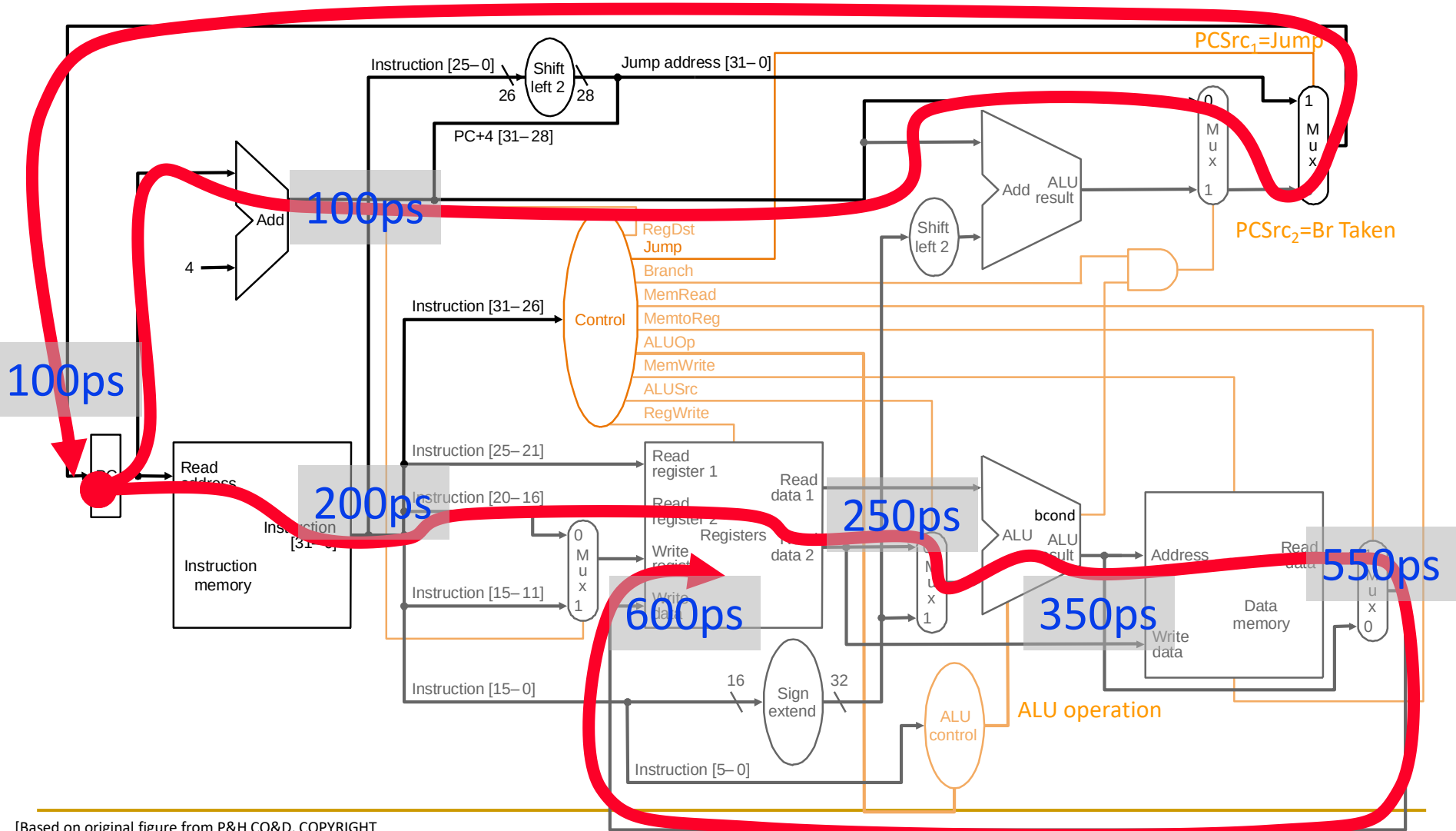
steps	IF	ID	EX	MEM	WB	Delay
resources	mem	RF	ALU	mem	RF	
R-type	200	50	100		50	400
I-type	200	50	100		50	400
LW	200	50	100	200	50	600
SW	200	50	100	200		550
Branch	200	50	100			350
Jump	200					200

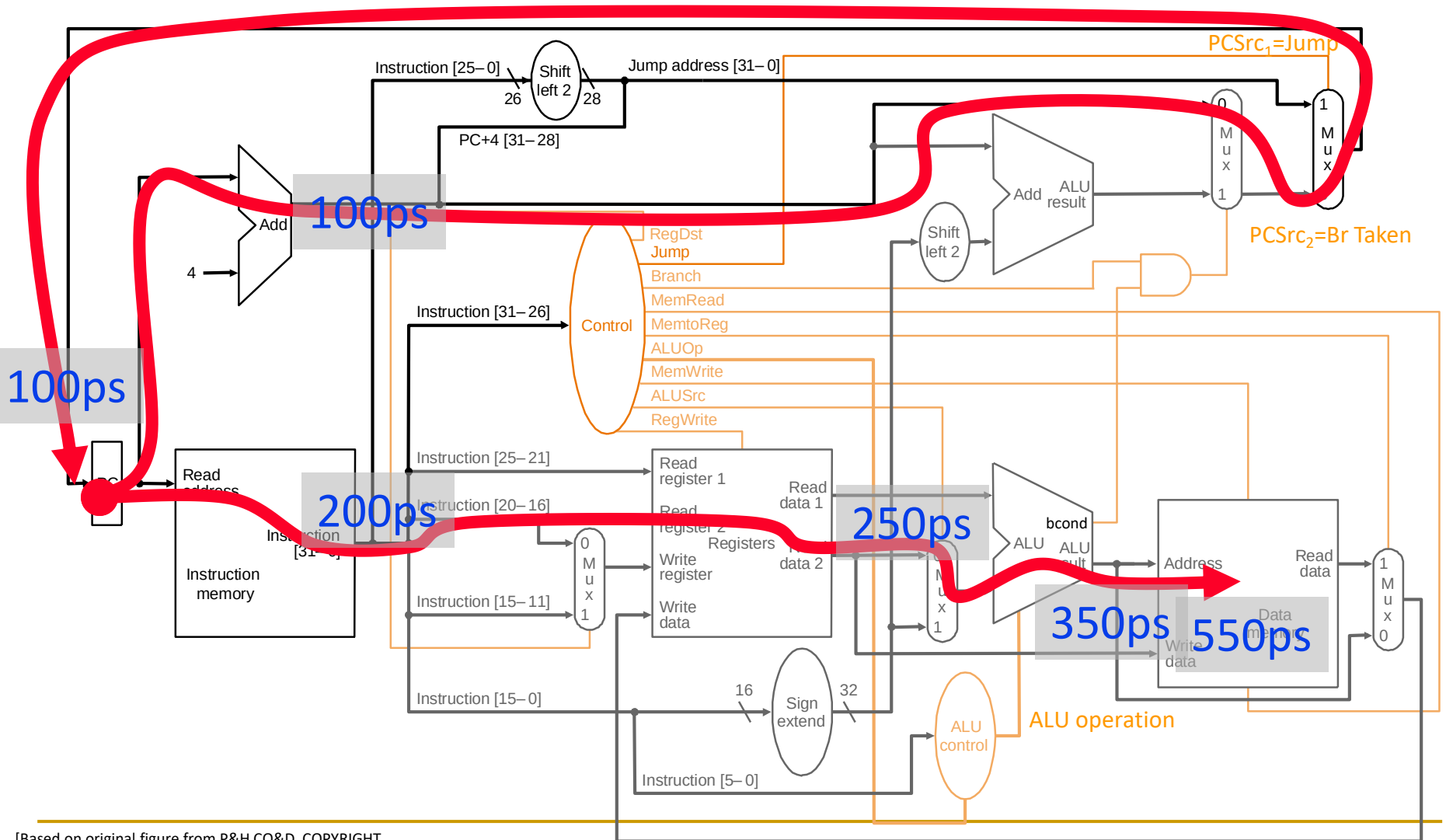
Let's Find the Critical Path



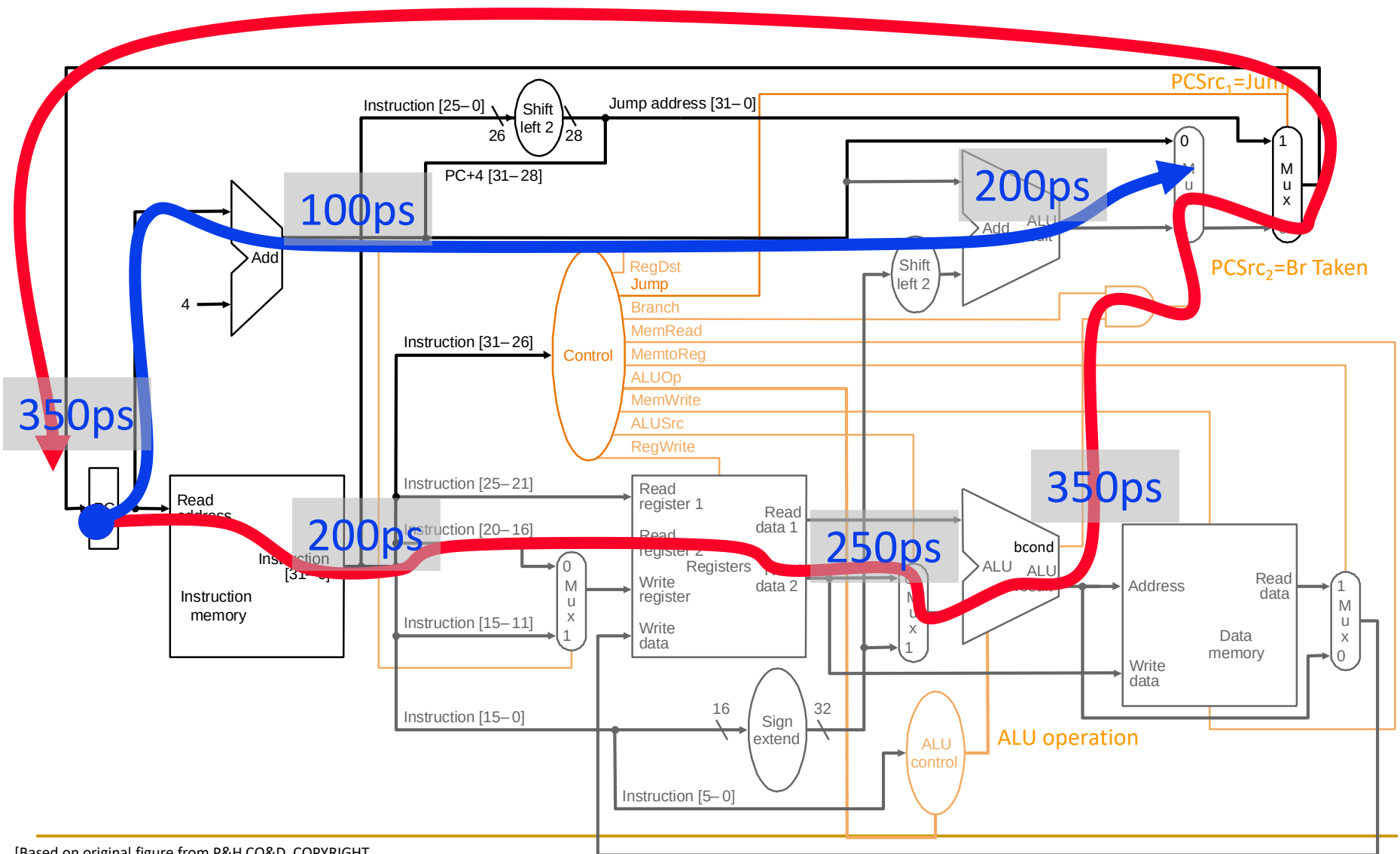
R-Type and I-Type ALU



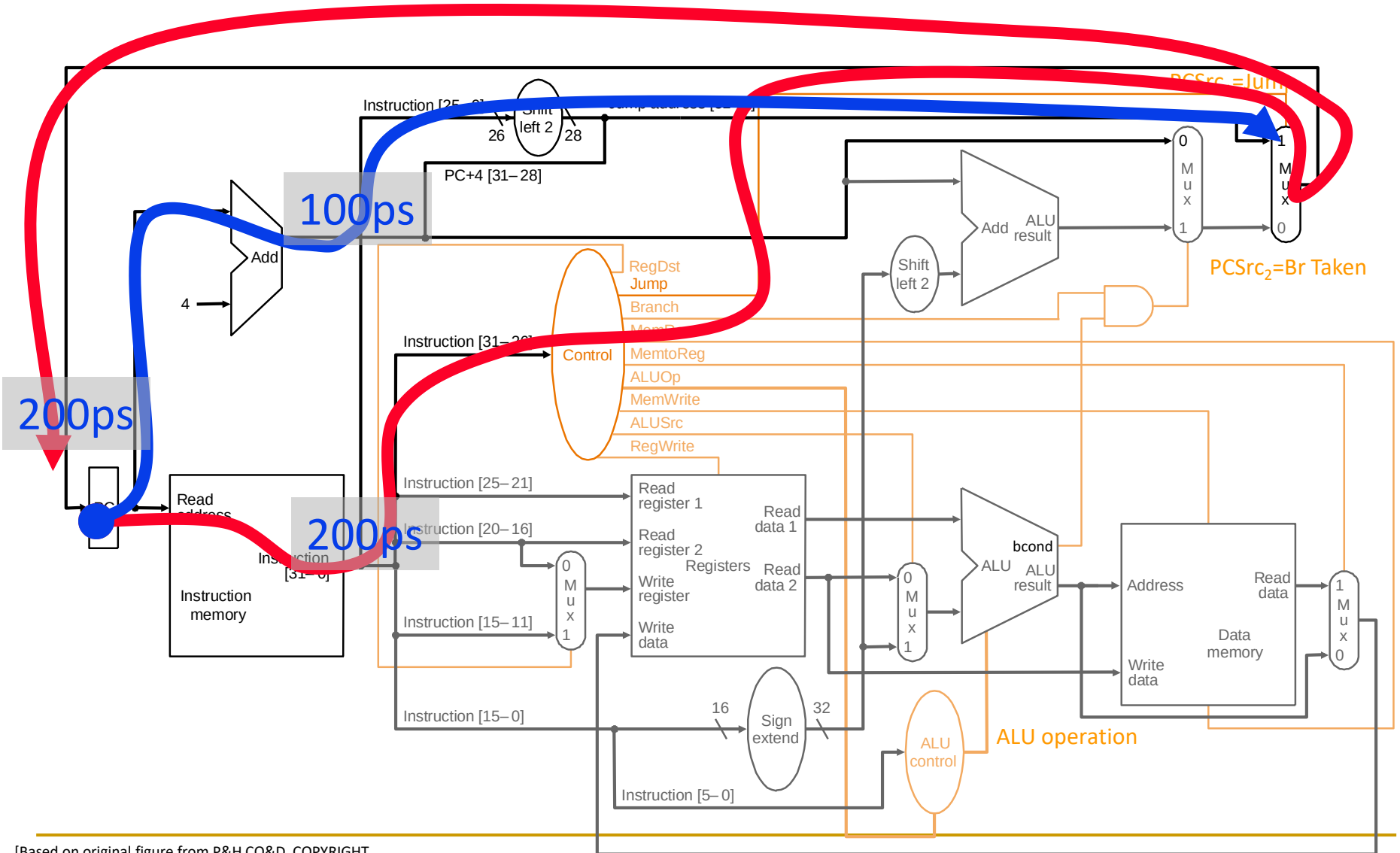




Branch Taken



Jump



Example Single-Cycle Datapath Analysis

- Assume (for the design in the previous slide)
 - ❑ memory units (read or write): 200 ps
 - ❑ ALU and adders: 100 ps
 - ❑ register file (read or write): 50 ps
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steps	IF	ID	EX	MEM	WB	Delay
resources	mem	RF	ALU	mem	RF	
R-type	200	50	100		50	400
I-type	200	50	100		50	400
LW	200	50	100	200	50	600
SW	200	50	100	200		550
Branch	200	50	100			350
Jump	200					200

What About Control Logic?

- How does that affect the critical path?
- Food for thought for you:
 - Can control logic be on the critical path?
 - Historical example:
 - CDC 5600: control store access took too long...

What is Really the Slowest Instruction to Process?

- Real world: **Memory is slow (not magic)**
- What if memory *sometimes* takes 150ns to access?
- Does it make sense to have a simple register to register add or jump to take {150ns + all else to perform a memory operation}?
- And, what if you need to access memory more than once to process an instruction?
 - Which instructions require this?

Single Cycle uArch: Complexity

■ Contrived

- All instructions run as slow as the slowest instruction

■ Inefficient

- All instructions run as slow as the slowest instruction
- Must provide worst-case combinational resources in parallel as required by any instruction
- Need to replicate a resource if it is needed more than once by an instruction during different parts of the instruction processing cycle

■ Not necessarily the simplest way to implement an ISA

- Tough for complex instructions, e.g., REP MOVSB (x86) or INDEX (VAX)

■ Not easy to optimize/improve performance

- Optimizing the common case (frequent instructions) does not work
- Need to optimize the worst case all the time

(Micro)architecture Design Principles

■ Critical path design

- Find and **decrease the maximum combinational logic delay**
- Break a path into multiple cycles if it takes too long

■ Bread and butter (common case) design

- **Spend time and resources on where it matters most**
 - i.e., improve what the machine is really designed to do
- Common case vs. uncommon case

■ Balanced design

- **Balance** instruction/data flow through hardware components
- **Design to eliminate bottlenecks:** balance the hardware for the work

Single-Cycle Design vs. Design Principles

- Critical path design
- Bread and butter (common case) design
- Balanced design

How does a single-cycle microarchitecture fare with respect to these principles?

Aside: System Design Principles

- When designing computer systems/architectures, it is important to follow good principles
 - Actually, this is true for **any system design**
 - Real architectures, buildings, bridges, train stations, ...
 - Good consumer products
 - Security & safety-critical systems
 - Decision making systems
 - ...
- Remember: “principled design” from our second lecture
 - Frank Lloyd Wright: “architecture [...] based upon **principle**, and not upon **precedent**”

Aside: Principled Architecture

- “architecture [...] based upon **principle**, and not upon **precedent**”



This



That



Recall: Takeaways

- It all starts from the basic building blocks and design principles
- And, knowledge of how to use, apply, enhance them
- Underlying technology might change (e.g., steel vs. wood)
 - but methods of taking advantage of technology bear resemblance
 - methods used for design depend on the principles employed

Aside: System Design Principles

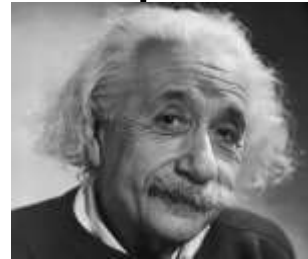
- We will continue to cover key principles in this course
- Here are some references where you can learn more
- Yale Patt, "Requirements, Bottlenecks, and Good Fortune: Agents for Microprocessor Evolution," Proc. of IEEE, 2001. (Levels of transformation, design point, etc)
- Mike Flynn, "Very High-Speed Computing Systems," Proc. of IEEE, 1966. (Flynn's Bottleneck → Balanced design)
- Gene M. Amdahl, "Validity of the single processor approach to achieving large scale computing capabilities," AFIPS Conference, April 1967. (Amdahl's Law → Common-case design)
- Butler W. Lampson, "Hints for Computer System Design," ACM Operating Systems Review, 1983.

A Key System Design Principle

- **Keep it simple**

- “Everything should be made as simple as possible, **but no simpler.**”

- Albert Einstein (paraphrased)



- And, **keep it low cost**: “An engineer is a person who can do for a dime what any fool can do for a dollar.”

- For more, see:

- Butler W. Lampson, “**Hints for Computer System Design**,” ACM Operating Systems Review, 1983, updated 2021.

- <https://arxiv.org/pdf/2011.02455>



Can We Do Better?

Multi-Cycle Microarchitectures

Multi-Cycle Microarchitectures

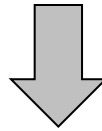
- Goal: Let each instruction take (close to) only as much time it really needs
- Idea
 - Determine clock cycle time independently of instruction processing time
 - Each instruction takes as many clock cycles as it needs to take
 - Multiple state transitions per instruction
 - The states followed by each instruction is different

Recall: The “Process Instruction” Step

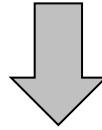
- ISA specifies abstractly what AS' should be, given an instruction and AS
 - It defines an **abstract finite state machine** where
 - State = programmer-visible state
 - Next-state logic = instruction execution specification
 - From ISA point of view, there are no “intermediate states” between AS and AS' during instruction execution
 - One state transition per instruction
- Microarchitecture implements how AS is transformed to AS'
 - There are many choices in implementation
 - We can have programmer-invisible state to optimize the speed of instruction execution: **multiple** state transitions per instruction
 - Choice 1: $AS \rightarrow AS'$ (transform AS to AS' in a single clock cycle)
 - Choice 2: $AS \rightarrow AS+MS1 \rightarrow AS+MS2 \rightarrow AS+MS3 \rightarrow AS'$ (take multiple clock cycles to transform AS to AS')

Multi-Cycle Microarchitecture

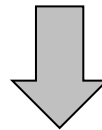
AS = Architectural (programmer visible) state
at the beginning of an instruction



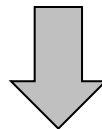
Step 1: Process part of instruction in one clock cycle



Step 2: Process part of instruction in the next clock cycle

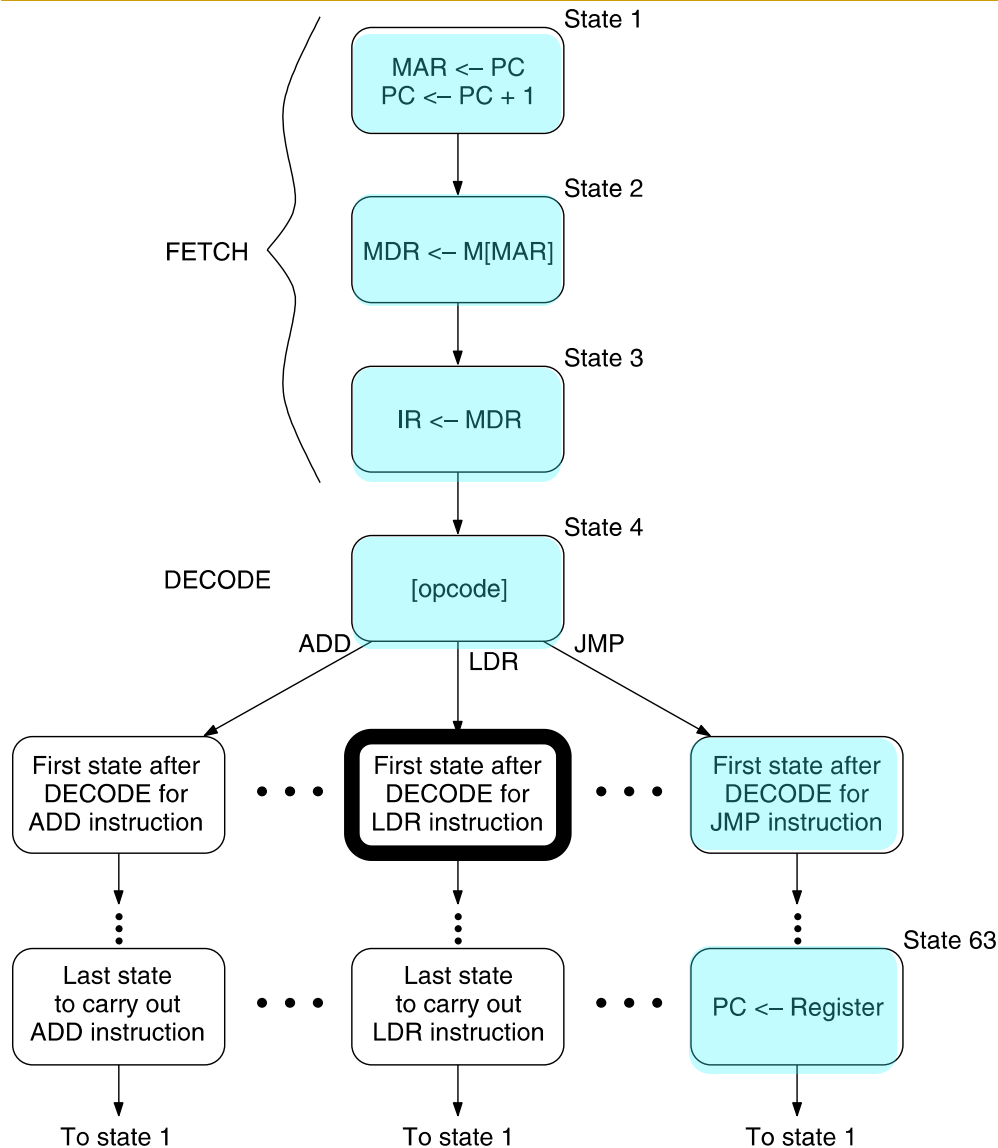


...



AS' = Architectural (programmer visible) state
at the end of a clock cycle

Recall: Control of the Instruction Cycle



- **State 1**
 - The FSM asserts GatePC and LD.MAR
 - It selects input (+1) in PCMUX and asserts LD.PC
- **State 2**
 - MDR is loaded with the instruction
- **State 3**
 - The FSM asserts GateMDR and LD.IR
- **State 4**
 - The FSM goes to next state depending on opcode
- **State 63**
 - JMP loads register into PC
- Full state diagram in Patt&Pattel, Appendix C

Figure 4.4 An abbreviated state diagram of the LC-3

This is an FSM Controlling a Multi-Cycle LC-3 Microarchitecture

Recall: Full State Machine for LC-3b

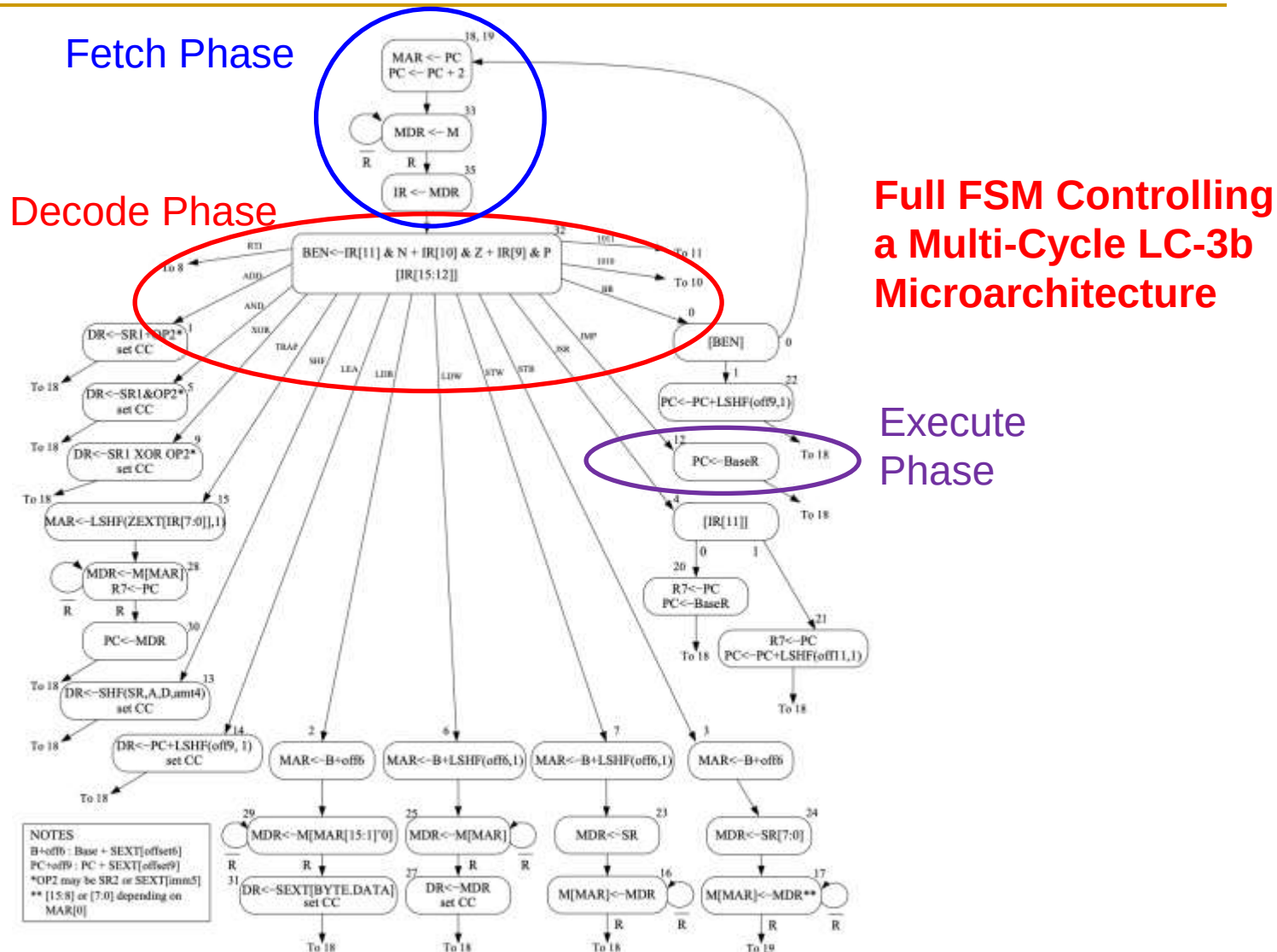


Figure C.2: A state machine for the LC-3b

Recall: LC-3 Multi-Cycle Implementation

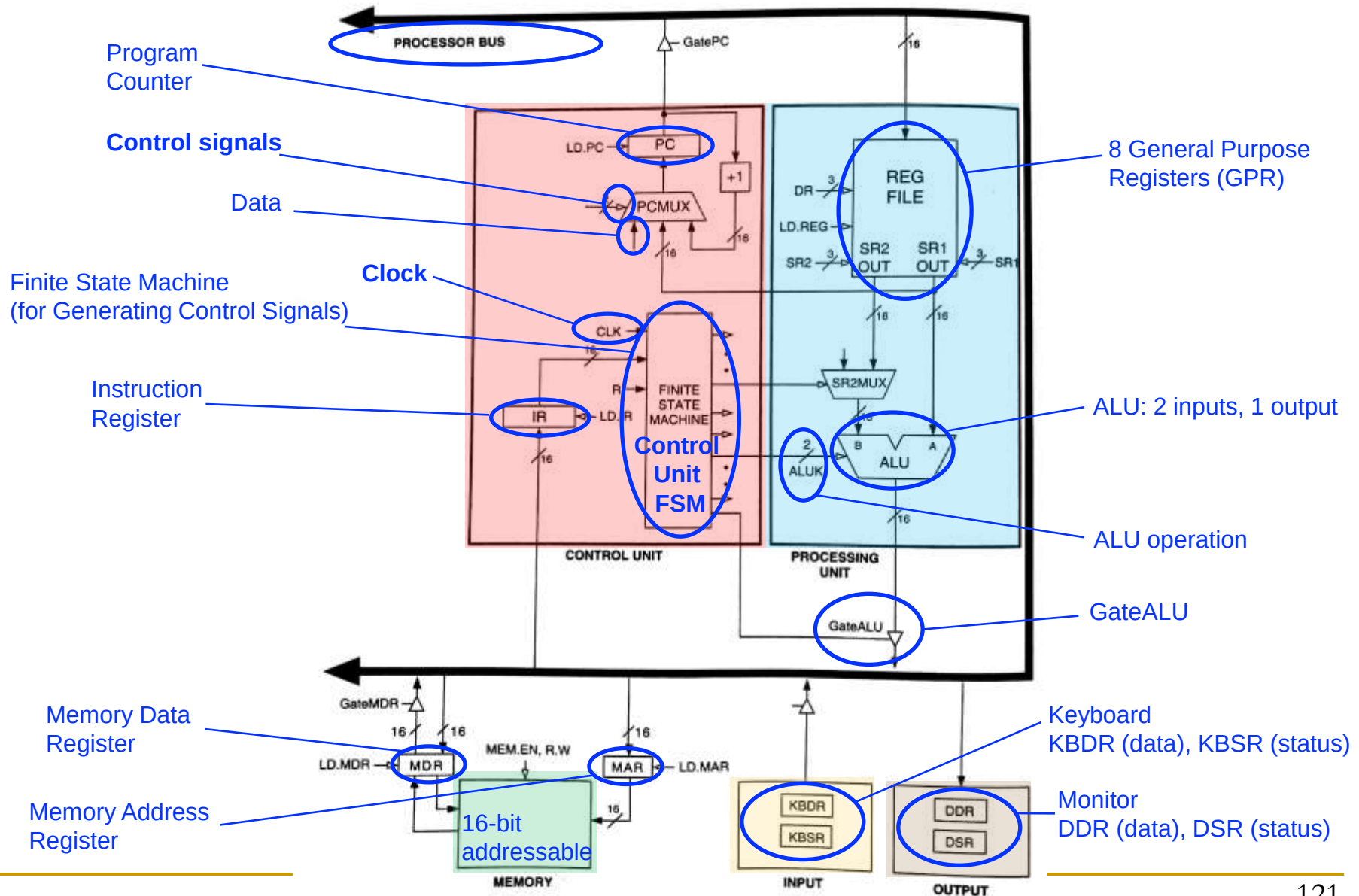


Figure 4.3 The LC-3 as an example of the von Neumann model

Benefits of Multi-Cycle Design

■ Critical path design

- ❑ Can keep reducing the critical path independently of the worst-case processing time of any instruction

■ Bread and butter (common case) design

- ❑ Can optimize the number of states it takes to execute “important” instructions that make up much of the execution time

■ Balanced design

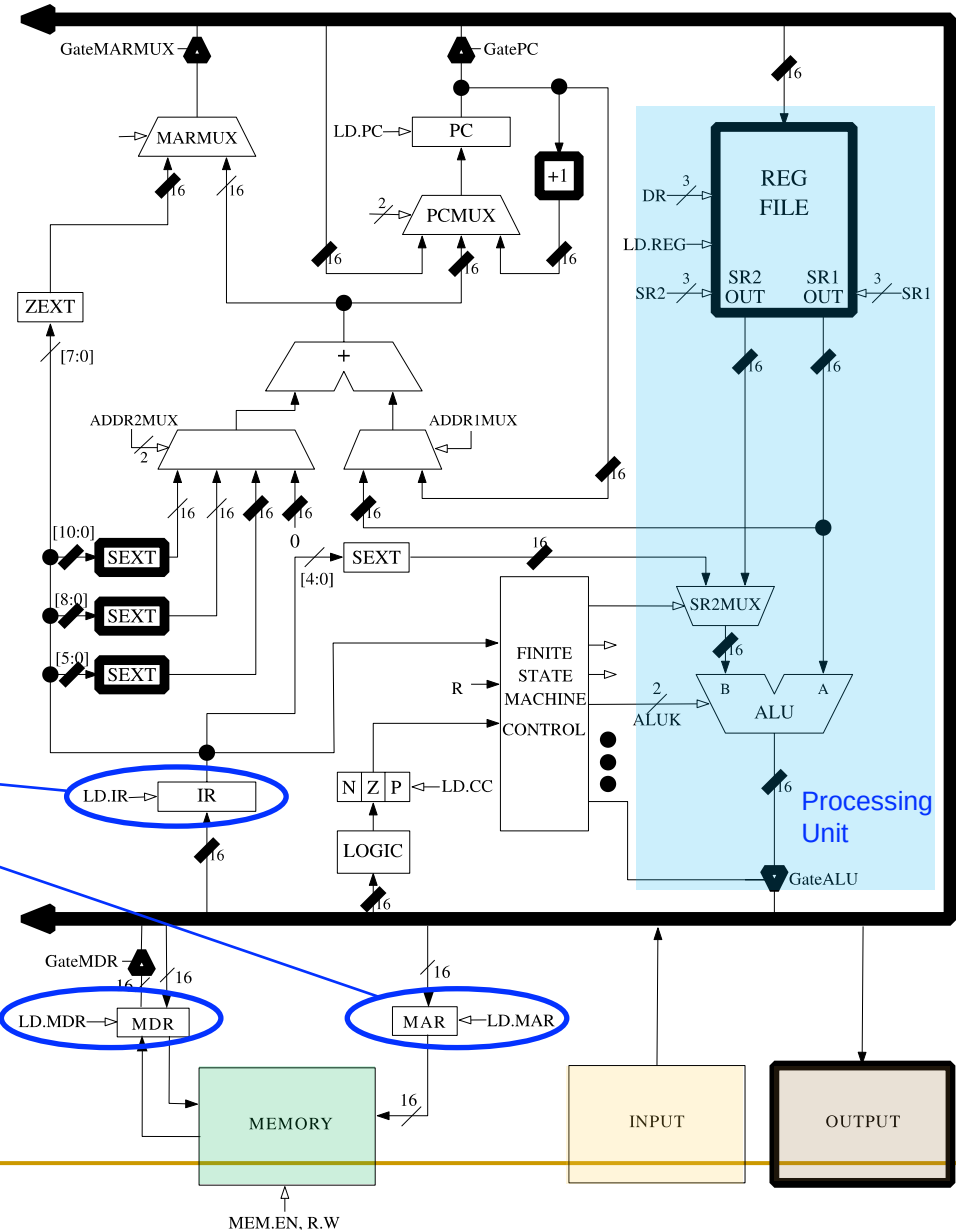
- ❑ No need to provide more capability or resources than really needed
 - An instruction that needs resource X multiple times does **not** require multiple X's to be implemented
 - Leads to more efficient hardware: Can reuse hardware components needed multiple times for an instruction

Downsides of Multi-Cycle Design

- **Need to store the intermediate results** at the end of each clock cycle
 - ❑ Hardware overhead for microarchitectural registers
 - ❑ Register setup/hold overhead (i.e., sequencing overhead) is paid multiple times for an instruction
- **Limited concurrency**
 - ❑ Only a small part of the machine is used at any point in time

Multi-Cycle LC-3 Data Path

Extra registers
not needed in a
single-cycle
design



Remember: Performance Analysis

- Execution time of a single instruction
 - **{CPI} x {clock cycle time}** CPI: Cycles Per Instruction
- Execution time of an entire program
 - Sum over all instructions [**{CPI} x {clock cycle time}**]
 - **{# of instructions} x {Average CPI} x {clock cycle time}**
- Single-cycle microarchitecture performance
 - $\text{CPI} = 1$
 - Clock cycle time = long
- Multi-cycle microarchitecture performance
 - CPI = different for each instruction
 - Average CPI $\rightarrow$ hopefully small
 - Clock cycle time = short

In multi-cycle, we have two degrees of freedom to optimize independently

Digital Design & Computer Arch.

Lecture 10: Microarchitecture Fundamentals & Design

Dr. Mohammad Sadrosadati

Prof. Onur Mutlu

ETH Zürich

Spring 2026

20 March 2026